1)

$I = \int \frac{1}{\sqrt{x^2+1}} \mathrm{d}x$

Use Integration by Substitution:

Let $x = \sinh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u \ \mathrm{d}u$

$\therefore I = \int \frac{1}{\sqrt{\sinh^2 u+1}} \times \cosh u \, \mathrm{d}u$

$= \int \frac{\cosh u}{\sqrt{\cosh^2 u}} \mathrm{d}u$

$= \int \frac{\cosh u}{\cosh u} \mathrm{d}u$

$= \int 1 \, \mathrm{d}u$

$= u + c$

$= \operatorname{arsinh} x + c$

2)

$I = \int \frac{1}{\sqrt{x^2-1}} \mathrm{d}x$

Use Integration by Substitution:

Let $x = \cosh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sinh u \Rightarrow \mathrm{d}x = \sinh u \, \mathrm{d}u$

$\therefore I = \int \frac{1}{\sqrt{\cosh^2 u-1}} \times \sinh u \, \mathrm{d}u$

$= \int \frac{\sinh u}{\sqrt{\sinh^2 u}} \mathrm{d}u$

$= \int \frac{\sinh u}{\sinh u} \mathrm{d}u$

$= \int 1 \, \mathrm{d}u$

$= u + c$

$= \operatorname{arcosh} x + c$

3)

$I = \int \frac{1}{1-x^2}\,\mathrm{d}x$

Use Integration by Substitution:

Let $x = \tanh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore I = \int \frac{1}{1-\tanh^2 u} \times \operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\,\mathrm{d}u$

$= \int 1\,\mathrm{d}u$

$= u + c$

$= \operatorname{artanh} x + c$

4) a)

$\int \frac{4}{\sqrt{x^2-1}}\,\mathrm{d}x$

$= 4\int \frac{1}{\sqrt{x^2-1}}\,\mathrm{d}x$

$= 4\operatorname{arcosh} x + c$

4) b)

$\int \frac{1}{2\sqrt{x^2+1}}\,\mathrm{d}x$

$= \frac{1}{2}\int \frac{1}{\sqrt{x^2+1}}\,\mathrm{d}x$

$= \frac{1}{2}\operatorname{arsinh} x + c$

4) c)

$\int \frac{6}{1-x^2}\,\mathrm{d}x$

$= 6\int \frac{1}{1-x^2}\,\mathrm{d}x$

$= 6\operatorname{artanh} x + c$

4) d)

$\int \frac{1}{\sqrt{4x^2-1}}\,\mathrm{d}x$

$= \int \frac{1}{\sqrt{(2x)^2-1}}\,\mathrm{d}x$

$= \frac{1}{2}\operatorname{arcosh} 2x + c$

4) e)

$$\int \frac{1}{\sqrt{9x^2+1}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{(3x)^2+1}}\,\mathrm{d}x$$
$$= \frac{1}{3}\operatorname{arsinh} 3x + c$$

4) f)

$$\int \frac{1}{1-25x^2}\,\mathrm{d}x$$
$$= 1\int \frac{1}{1-(5x)^2}\,\mathrm{d}x$$
$$= \frac{1}{5}\operatorname{artanh} 5x + c$$

4) g)

$$\int \frac{10}{\sqrt{16x^2-1}}\,\mathrm{d}x$$
$$= \int \frac{10}{\sqrt{(4x)^2-1}}\,\mathrm{d}x$$
$$= 10 \times \frac{1}{4} \times \operatorname{arcosh} 4x + c$$
$$= \frac{5}{2}\operatorname{arcosh} 4x + c$$

4) h)

$$\int \frac{\sqrt{8}}{\sqrt{2x^2+1}}\,\mathrm{d}x$$
$$= \int \frac{2\sqrt{2}}{\sqrt{\left(\sqrt{2}x\right)^2+1}}\,\mathrm{d}x$$
$$= 2\sqrt{2} \times \frac{1}{\sqrt{2}} \times \operatorname{arsinh}\left(\sqrt{2}x\right) + c$$
$$= 2\operatorname{arsinh}\left(\sqrt{2}x\right) + c$$

4) i)

$$\int \frac{100}{1-\frac{1}{5}x^2}\,\mathrm{d}x$$

$$= \int \frac{100}{1-\left(\frac{1}{\sqrt{5}}x\right)^2}\,\mathrm{d}x$$

$$= 100 \times \frac{1}{\frac{1}{\sqrt{5}}} \times \operatorname{artanh}\left(\frac{1}{\sqrt{5}}x\right) + c$$

$$= 100\sqrt{5}\operatorname{artanh}\left(\frac{\sqrt{5}}{5}x\right) + c$$

5) a)

$$\frac{1}{15-x^2} \equiv \frac{1}{15\left(1-\frac{1}{15}x^2\right)} \equiv \frac{1}{15\left(1-\left(\frac{1}{\sqrt{15}}x\right)^2\right)}$$

So $k = \frac{1}{\sqrt{15}} = \frac{\sqrt{15}}{15}$

5) b)

$$\int \frac{1}{15-x^2}\,\mathrm{d}x = \int \frac{1}{15\left(1-\left(\frac{1}{\sqrt{15}}x\right)^2\right)}\,\mathrm{d}x$$

$$= \frac{1}{15}\int \frac{1}{1-\left(\frac{1}{\sqrt{15}}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{15} \times \frac{1}{\frac{1}{\sqrt{15}}} \times \operatorname{artanh}\left(\frac{1}{\sqrt{15}}x\right) + c$$

$$= \frac{\sqrt{15}}{15}\operatorname{artanh}\left(\frac{\sqrt{15}}{15}x\right) + c$$

6) a)

$$\frac{1}{\sqrt{x^2+3}} \equiv \frac{1}{\sqrt{3\left(\frac{1}{3}x^2+1\right)}} \equiv \frac{1}{\sqrt{3}\sqrt{\left(\frac{1}{\sqrt{3}}x\right)^2+1}}$$

So $k = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

6) b)

$$\int \frac{1}{\sqrt{x^2+3}}\,\mathrm{d}x = \int \frac{1}{\sqrt{3}\sqrt{\left(\frac{1}{\sqrt{3}}x\right)^2+1}}\,\mathrm{d}x$$

$$= \frac{1}{\sqrt{3}}\int \frac{1}{\sqrt{\left(\frac{1}{\sqrt{3}}x\right)^2+1}}\,\mathrm{d}x$$

$$= \frac{1}{\sqrt{3}} \times \frac{1}{\frac{1}{\sqrt{3}}} \times \operatorname{arsinh}\left(\frac{1}{\sqrt{3}}x\right) + c$$

$$= \operatorname{arsinh}\left(\frac{\sqrt{3}}{3}x\right) + c$$

7) a)

$$\int \frac{1}{25-x^2}\,\mathrm{d}x$$

$$= \int \frac{1}{25\left(1-\frac{1}{25}x^2\right)}\,\mathrm{d}x$$

$$= \frac{1}{25}\int \frac{1}{1-\left(\frac{1}{5}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{25} \times \frac{1}{\frac{1}{5}} \times \operatorname{artanh}\left(\frac{1}{5}x\right) + c$$

$$= \frac{1}{5}\operatorname{artanh}\left(\frac{1}{5}x\right) + c$$

7) b)

$$\int \frac{1}{\sqrt{x^2+9}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{9\left(\frac{1}{9}x^2+1\right)}}\,\mathrm{d}x$$

$$= \frac{1}{3}\int \frac{1}{\sqrt{\left(\frac{1}{3}x\right)^2+1}}\,\mathrm{d}x$$

$$= \frac{1}{3} \times \frac{1}{\frac{1}{3}} \times \operatorname{arsinh}\left(\frac{1}{3}x\right) + c$$

$$= \operatorname{arsinh}\left(\frac{1}{3}x\right) + c$$

7) c)

$$\int \frac{1}{\sqrt{x^2-4}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{4\left(\frac{1}{4}x^2-1\right)}}\,\mathrm{d}x$$
$$= \frac{1}{2}\int \frac{1}{\sqrt{\left(\frac{1}{2}x\right)^2-1}}\,\mathrm{d}x$$
$$= \frac{1}{2}\times\frac{1}{\frac{1}{2}}\times \operatorname{arcosh}\left(\frac{1}{2}x\right)+c$$
$$= \operatorname{arcosh}\left(\frac{1}{2}x\right)+c$$

7) d)

$$\int \frac{1}{8-x^2}\,\mathrm{d}x$$
$$= \int \frac{1}{8\left(1-\frac{1}{8}x^2\right)}\,\mathrm{d}x$$
$$= \frac{1}{8}\int \frac{1}{1-\left(\frac{1}{\sqrt{8}}x\right)^2}\,\mathrm{d}x$$
$$= \frac{1}{8}\times\frac{1}{\frac{1}{\sqrt{8}}}\times \operatorname{artanh}\left(\frac{1}{2\sqrt{2}}x\right)+c$$
$$= \frac{\sqrt{2}}{4}\operatorname{artanh}\left(\frac{\sqrt{2}}{4}x\right)+c$$

7) e)

$$\int \frac{1}{\sqrt{x^2+12}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{12\left(\frac{1}{12}x^2+1\right)}}\,\mathrm{d}x$$
$$= \frac{1}{\sqrt{12}}\int \frac{1}{\sqrt{\left(\frac{1}{\sqrt{12}}x\right)^2+1}}\,\mathrm{d}x$$
$$= \frac{1}{\sqrt{12}}\times\frac{1}{\frac{1}{\sqrt{12}}}\times \operatorname{arsinh}\left(\frac{1}{\sqrt{12}}x\right)+c$$
$$= \operatorname{arsinh}\left(\frac{\sqrt{3}}{6}x\right)+c$$

7) f)

$$\int \frac{1}{\sqrt{x^2-20}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{20\left(\frac{1}{20}x^2-1\right)}}\,\mathrm{d}x$$
$$= \frac{1}{\sqrt{20}}\int \frac{1}{\sqrt{\left(\frac{1}{\sqrt{20}}x\right)^2-1}}\,\mathrm{d}x$$
$$= \frac{1}{\sqrt{20}} \times \frac{1}{\frac{1}{\sqrt{20}}} \times \operatorname{arcosh}\left(\frac{1}{\sqrt{20}}x\right) + c$$
$$= \operatorname{arcosh}\left(\frac{\sqrt{5}}{10}x\right) + c$$

7) g)

$$\int \frac{5}{10-x^2}\,\mathrm{d}x$$
$$= \int \frac{5}{10\left(1-\frac{1}{10}x^2\right)}\,\mathrm{d}x$$
$$= \frac{1}{2}\int \frac{1}{1-\left(\frac{1}{\sqrt{10}}x\right)^2}\,\mathrm{d}x$$
$$= \frac{1}{2} \times \frac{1}{\frac{1}{\sqrt{10}}} \times \operatorname{artanh}\left(\frac{1}{\sqrt{10}}x\right) + c$$
$$= \frac{\sqrt{10}}{2}\operatorname{artanh}\left(\frac{\sqrt{10}}{10}x\right) + c$$

7) h)

$$\int \frac{14}{\sqrt{x^2+7}}\,\mathrm{d}x$$
$$= \int \frac{14}{\sqrt{7\left(\frac{1}{7}x^2+1\right)}}\,\mathrm{d}x$$
$$= \frac{14}{\sqrt{7}}\int \frac{1}{\sqrt{\left(\frac{1}{\sqrt{7}}x\right)^2+1}}\,\mathrm{d}x$$
$$= \frac{14}{\sqrt{7}} \times \frac{1}{\frac{1}{\sqrt{7}}} \times \operatorname{arsinh}\left(\frac{1}{\sqrt{7}}x\right) + c$$
$$= 14\operatorname{arsinh}\left(\frac{\sqrt{7}}{7}x\right) + c$$

7) i)

$$\int \frac{18}{\sqrt{x^2-3}}\,\mathrm{d}x$$
$$= \int \frac{18}{\sqrt{3\left(\frac{1}{3}x^2-1\right)}}\,\mathrm{d}x$$
$$= \frac{18}{\sqrt{3}}\int \frac{1}{\sqrt{\left(\frac{1}{\sqrt{3}}x\right)^2-1}}\,\mathrm{d}x$$
$$= \frac{18}{\sqrt{3}} \times \frac{1}{\frac{1}{\sqrt{3}}} \times \operatorname{arcosh}\left(\frac{1}{\sqrt{3}}x\right) + c$$
$$= 18\operatorname{arcosh}\left(\frac{\sqrt{3}}{3}x\right) + c$$

7) j)

$$\int \frac{1}{9-4x^2}\,\mathrm{d}x$$
$$= \int \frac{1}{9\left(1-\frac{4}{9}x^2\right)}\,\mathrm{d}x$$
$$= \frac{1}{9}\int \frac{1}{1-\left(\frac{2}{3}x\right)^2}\,\mathrm{d}x$$
$$= \frac{1}{9} \times \frac{1}{\frac{2}{3}} \times \operatorname{artanh}\left(\frac{2}{3}x\right) + c$$
$$= \frac{1}{6}\operatorname{artanh}\left(\frac{2}{3}x\right) + c$$

7) k)

$$\int \frac{1}{\sqrt{9x^2+4}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{4\left(\frac{9}{4}x^2+1\right)}}\,\mathrm{d}x$$
$$= \frac{1}{2}\int \frac{1}{\sqrt{\left(\frac{3}{2}x\right)^2+1}}\,\mathrm{d}x$$
$$= \frac{1}{2} \times \frac{1}{\frac{3}{2}} \times \operatorname{arsinh}\left(\frac{3}{2}x\right) + c$$
$$= \frac{1}{3}\operatorname{arsinh}\left(\frac{3}{2}x\right) + c$$

7) l)

$$\int \frac{1}{\sqrt{25x^2-16}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{16\left(\frac{25}{16}x^2-1\right)}}\,\mathrm{d}x$$
$$= \frac{1}{4}\int \frac{1}{\sqrt{\left(\frac{5}{4}x\right)^2-1}}\,\mathrm{d}x$$
$$= \frac{1}{4} \times \frac{1}{\frac{5}{4}} \times \operatorname{arcosh}\left(\frac{5}{4}x\right) + c$$
$$= \frac{1}{5}\operatorname{arcosh}\left(\frac{5}{4}x\right) + c$$

7) m)

$$\int \frac{15}{4-\frac{1}{4}x^2}\,\mathrm{d}x$$
$$= \int \frac{15}{4\left(1-\frac{1}{16}x^2\right)}\,\mathrm{d}x$$
$$= \frac{15}{4}\int \frac{1}{1-\left(\frac{1}{4}x\right)^2}\,\mathrm{d}x$$
$$= \frac{15}{4} \times \frac{1}{\frac{1}{4}} \times \operatorname{artanh}\left(\frac{1}{4}x\right) + c$$
$$= 15\operatorname{artanh}\left(\frac{1}{4}x\right) + c$$

7) n)

$$\int \frac{3}{\sqrt{16x^2+\frac{1}{9}}}\,\mathrm{d}x$$
$$= \int \frac{3}{\sqrt{\frac{1}{9}(144x^2+1)}}\,\mathrm{d}x$$
$$= \frac{3}{\frac{1}{3}}\int \frac{1}{\sqrt{(12x)^2+1}}\,\mathrm{d}x$$
$$= 9 \times \frac{1}{12} \times \operatorname{arsinh}(12x) + c$$
$$= \frac{3}{4}\operatorname{arsinh}(12x) + c$$

7) o)

$$\int \frac{18}{\sqrt{\frac{4}{9}x^2-\frac{9}{4}}}\,\mathrm{d}x$$
$$= \int \frac{18}{\sqrt{\frac{9}{4}\left(\frac{16}{81}x^2-1\right)}}\,\mathrm{d}x$$
$$= \frac{18}{\frac{3}{2}}\int \frac{1}{\sqrt{\left(\frac{4}{9}x\right)^2-1}}\,\mathrm{d}x$$
$$= 12 \times \frac{1}{\frac{4}{9}} \times \operatorname{arcosh}\left(\frac{4}{9}x\right) + c$$
$$= 27 \operatorname{arcosh}\left(\frac{4}{9}x\right) + c$$

8) a)

$$\int \frac{1}{a^2-x^2}\,\mathrm{d}x$$
$$= \int \frac{1}{a^2\left(1-\frac{1}{a^2}x^2\right)}\,\mathrm{d}x$$
$$= \frac{1}{a^2}\int \frac{1}{1-\left(\frac{1}{a}x\right)^2}\,\mathrm{d}x$$
$$= \frac{1}{a^2} \times \frac{1}{\frac{1}{a}} \times \operatorname{artanh}\left(\frac{1}{a}x\right) + c$$
$$= \frac{1}{a}\operatorname{artanh}\left(\frac{1}{a}x\right) + c$$

8) b)

$$\int \frac{1}{\sqrt{x^2+a^2}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{\frac{1}{a^2}(a^2x^2+1)}}\,\mathrm{d}x$$
$$= \frac{1}{\frac{1}{a}}\int \frac{1}{\sqrt{(ax)^2+1}}\,\mathrm{d}x$$
$$= a \times \frac{1}{a} \times \operatorname{arsinh}\left(\frac{1}{a}x\right) + c$$
$$= \operatorname{arsinh}\left(\frac{1}{a}x\right) + c$$

8) c)

$$\int \frac{1}{\sqrt{k^2x^2-a^2}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{a^2\left(\frac{k^2}{a^2}x^2-1\right)}}\,\mathrm{d}x$$
$$= \frac{1}{a}\int \frac{1}{\sqrt{\left(\frac{k}{a}x\right)^2-1}}\,\mathrm{d}x$$
$$= \frac{1}{a} \times \frac{1}{\frac{k}{a}} \times \operatorname{arcosh}\left(\frac{k}{a}x\right) + c$$
$$= \frac{1}{k}\operatorname{arcosh}\left(\frac{k}{a}x\right) + c$$

9) a)

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{1-x^2}\,\mathrm{d}x$$
$$= [\operatorname{artanh} x]_{-\frac{1}{2}}^{\frac{1}{2}}$$
$$= \left(\operatorname{artanh}\left(\frac{1}{2}\right)\right) - \left(\operatorname{artanh}\left(-\frac{1}{2}\right)\right)$$
$$= \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) - \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)}\right)$$
$$= \frac{1}{2}\ln 3 - \frac{1}{2}\ln\frac{1}{3}$$
$$= \frac{1}{2}\ln 3 - \frac{1}{2}\ln 3^{-1}$$
$$= \frac{1}{2}\ln 3 + \frac{1}{2}\ln 3$$
$$= \ln 3$$

9) b)

$$\int_1^2 \frac{1}{\sqrt{4x^2-1}}\,\mathrm{d}x$$
$$= \int_1^2 \frac{1}{\sqrt{(2x)^2-1}}\,\mathrm{d}x$$
$$= \left[\frac{1}{2}\operatorname{arcosh} 2x\right]_1^2$$
$$= \left(\frac{1}{2}\operatorname{arcosh}(2\times 2)\right) - \left(\frac{1}{2}\operatorname{arcosh}(2\times 1)\right)$$
$$= \frac{1}{2}\operatorname{arcosh} 4 - \frac{1}{2}\operatorname{arcosh} 2$$
$$= \frac{1}{2}\ln\left(4+\sqrt{4^2-1}\right) - \frac{1}{2}\ln\left(2+\sqrt{2^2-1}\right)$$
$$= \frac{1}{2}\ln\left(4+\sqrt{15}\right) - \frac{1}{2}\ln\left(2+\sqrt{3}\right)$$
$$= \frac{1}{2}\left(\ln\left(4+\sqrt{15}\right) - \ln\left(2+\sqrt{3}\right)\right)$$
$$= \frac{1}{2}\ln\left(\frac{4+\sqrt{15}}{2+\sqrt{3}}\right)$$

9) c)

$$\int_{-5}^5 \frac{30}{\sqrt{36x^2+25}}\,\mathrm{d}x$$
$$= \int_{-5}^5 \frac{30}{\sqrt{25\left(\frac{36}{25}x^2+1\right)}}\,\mathrm{d}x$$
$$= \frac{30}{5}\int_{-5}^5 \frac{1}{\sqrt{\left(\frac{6}{5}x\right)^2+1}}\,\mathrm{d}x$$
$$= 6\left[\frac{1}{\frac{6}{5}}\operatorname{arsinh}\left(\frac{6}{5}x\right)\right]_{-5}^5$$
$$= \left[5\operatorname{arsinh}\left(\frac{6}{5}x\right)\right]_{-5}^5$$
$$= \left(5\operatorname{arsinh}\left(\frac{6}{5}\times 5\right)\right) - \left(5\operatorname{arsinh}\left(\frac{6}{5}\times(-5)\right)\right)$$
$$= 5\operatorname{arsinh} 6 - 5\operatorname{arsinh}(-6)$$
$$= 5\operatorname{arsinh} 6 + 5\operatorname{arsinh} 6$$
$$= 10\operatorname{arsinh} 6$$
$$= 10\ln\left(6+\sqrt{6^2+1}\right)$$
$$= 10\ln\left(6+\sqrt{37}\right)$$

10)

$$\int_{-1}^{1} \frac{3}{9-4x^2}\,\mathrm{d}x$$

$$= \int_{-1}^{1} \frac{3}{9\left(1-\frac{4}{9}x^2\right)}\,\mathrm{d}x$$

$$= \frac{1}{3}\int_{-1}^{1} \frac{1}{1-\left(\frac{2}{3}x\right)^2}\,\mathrm{d}x$$

$$= \frac{1}{3}\left[\frac{1}{\frac{2}{3}} \times \operatorname{artanh}\left(\frac{2}{3}x\right)\right]_{-1}^{1}$$

$$= \left[\frac{1}{2}\operatorname{artanh}\left(\frac{2}{3}x\right)\right]_{-1}^{1}$$

$$= \left(\frac{1}{2}\operatorname{artanh}\left(\frac{2}{3}\right)\right) - \left(\frac{1}{2}\operatorname{artanh}\left(-\frac{2}{3}\right)\right)$$

$$= \frac{1}{2} \times \frac{1}{2}\ln\left(\frac{1+\frac{2}{3}}{1-\frac{2}{3}}\right) - \frac{1}{2} \times \frac{1}{2}\ln\left(\frac{1+\left(-\frac{2}{3}\right)}{1-\left(-\frac{2}{3}\right)}\right)$$

$$= \frac{1}{4}\ln 5 - \frac{1}{4}\ln\frac{1}{5}$$

$$= \frac{1}{4}\ln 5 - \frac{1}{4}\ln(5^{-1})$$

$$= \frac{1}{4}\ln 5 + \frac{1}{4}\ln 5$$

$$= \frac{1}{2}\ln 5$$

11)

$$\int_{\frac{3}{5}}^{\frac{4}{5}} \left(\frac{5}{\sqrt{25x^2-4}} + \frac{31}{155x-92} \right) \mathrm{d}x$$

$$= \int_{\frac{3}{5}}^{\frac{4}{5}} \left(\frac{5}{\sqrt{4\left(\frac{25}{4}x^2-1\right)}} + \frac{31}{155x-92} \right) \mathrm{d}x$$

$$= \int_{\frac{3}{5}}^{\frac{4}{5}} \left(\frac{5}{2} \times \frac{1}{\sqrt{\left(\frac{5}{2}x\right)^2-1}} + \frac{31}{155x-92} \right) \mathrm{d}x$$

$$= \left[\frac{5}{2} \times \frac{1}{\frac{5}{2}} \times \operatorname{arcosh}\left(\frac{5}{2}x\right) + \frac{1}{5}\ln|155x-92| \right]_{\frac{3}{5}}^{\frac{4}{5}}$$

$$= \left[\operatorname{arcosh}\left(\frac{5}{2}x\right) + \frac{1}{5}\ln|155x-92| \right]_{\frac{3}{5}}^{\frac{4}{5}}$$

$$= \left(\operatorname{arcosh} 2 + \frac{1}{5}\ln 32 \right) - \left(\operatorname{arcosh}\frac{3}{2} + \frac{1}{5}\ln 1 \right)$$

$$= \left(\ln\left(2+\sqrt{2^2-1}\right) + \ln 32^{\frac{1}{5}} \right) - \left(\ln\left(\frac{3}{2}+\sqrt{\left(\frac{3}{2}\right)^2-1}\right) + 0 \right)$$

$$= \ln\left(2+\sqrt{3}\right) + \ln 2 - \ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)$$

$$= \ln\left(4+2\sqrt{3}\right) - \ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)$$

$$= \ln\left(\frac{4+2\sqrt{3}}{\frac{3}{2}+\frac{\sqrt{5}}{2}}\right)$$

$$= \ln\left(\frac{8+4\sqrt{3}}{3+\sqrt{5}}\right)$$

$$= \ln\left(\frac{8+4\sqrt{3}}{3+\sqrt{5}} \times \frac{3-\sqrt{5}}{3-\sqrt{5}}\right)$$

$$= \ln\left(\frac{24-8\sqrt{5}+12\sqrt{3}-4\sqrt{15}}{4}\right)$$

$$= \ln\left(6-2\sqrt{5}+3\sqrt{3}-\sqrt{15}\right)$$

12)

$\int_1^{\frac{3}{2}} \frac{4}{\sqrt{4x^2-1}}\,\mathrm{d}x$

$= \int_1^{\frac{3}{2}} \frac{4}{\sqrt{(2x)^2-1}}\,\mathrm{d}x$

$= \left[4 \times \frac{1}{2}\operatorname{arcosh} 2x\right]_1^{\frac{3}{2}}$

$= [2\operatorname{arcosh} 2x]_1^{\frac{3}{2}}$

$= (2\operatorname{arcosh} 3) - (2\operatorname{arcosh} 2)$

$= 2\ln\left(3+\sqrt{3^2-1}\right) - 2\ln\left(2+\sqrt{2^2-1}\right)$

$= 2\ln\left(3+2\sqrt{2}\right) - 2\ln\left(2+\sqrt{3}\right)$

$= \ln\left(\left(3+2\sqrt{2}\right)^2\right) - \ln\left(\left(2+\sqrt{3}\right)^2\right)$

$= \ln\left(17+12\sqrt{2}\right) - \ln\left(7+4\sqrt{3}\right)$

$= \ln\left(\frac{17+12\sqrt{2}}{7+4\sqrt{3}}\right)$

$= \ln\left(119-48\sqrt{6}-68\sqrt{3}+84\sqrt{2}\right)$

So $a = 15$, $b = 5$, $c = 3$ and $d = 84$

13) a)

$\frac{19-39x^2}{(1-x^2)(4-9x^2)} \equiv \frac{Ax+B}{1-x^2} + \frac{Cx+D}{4-9x^2}$

$\Rightarrow 19-39x^2 \equiv (Ax+B)(4-9x^2) + (Cx+D)(1-x^2)$

Let $x = 1$: $-20 = (A+B) \times (-5) \Rightarrow A+B = 4$

Let $x = -1$: $-20 = (-A+B) \times (-5) \Rightarrow -A+B = 4$

Let $x = \frac{2}{3}$: $\frac{5}{3} = \left(\frac{2}{3}C+D\right) \times \frac{5}{9} \Rightarrow \frac{2}{3}C+D = 3$

Let $x = -\frac{2}{3}$: $\frac{5}{3} = \left(-\frac{2}{3}C+D\right) \times \frac{5}{9} \Rightarrow -\frac{2}{3}C+D = 3$

$A = 0$, $B = 4$, $C = 0$, $D = 3$

$\therefore \frac{19-39x^2}{(1-x^2)(4-9x^2)} \equiv \frac{4}{1-x^2} + \frac{3}{4-9x^2}$

13) b)

$\int \frac{19-39x^2}{(1-x^2)(4-9x^2)}\,\mathrm{d}x = \int \left(\frac{4}{1-x^2}+\frac{3}{4-9x^2}\right)\mathrm{d}x$

$= \int \left(\frac{4}{1-x^2}+\frac{3}{4\left(1-\frac{9}{4}x^2\right)}\right)\mathrm{d}x$

$= \int \left(\frac{4}{1-x^2}+\frac{3}{4}\times\frac{1}{1-\left(\frac{3}{2}x\right)^2}\right)\mathrm{d}x$

$= 4\operatorname{artanh} x + \frac{3}{4}\times\frac{1}{\frac{3}{2}}\times\operatorname{artanh}\left(\frac{3}{2}x\right)+c$

$= 4\operatorname{artanh} x + \frac{1}{2}\operatorname{artanh}\left(\frac{3}{2}x\right)+c$

$\therefore p=4,\ q=1,\ r=\frac{1}{2},\ s=\frac{3}{2}$

14) a)

$\frac{12x^2+1}{(1-9x^2)(1-16x^2)} \equiv \frac{Ax+B}{1-9x^2}+\frac{Cx+D}{1-16x^2}$

$\Rightarrow 12x^2+1 \equiv (Ax+B)(1-16x^2)+(Cx+D)(1-9x^2)$

Let $x=\frac{1}{3}: \frac{7}{3} = \left(\frac{1}{3}A+B\right)\times\left(-\frac{7}{9}\right) \Rightarrow \frac{1}{3}A+B=-3$

Let $x=-\frac{1}{3}: \frac{7}{3} = \left(-\frac{1}{3}A+B\right)\times\left(-\frac{7}{9}\right) \Rightarrow -\frac{1}{3}A+B=-3$

Let $x=\frac{1}{4}: \frac{7}{4} = \left(\frac{1}{4}C+D\right)\times\frac{7}{16} \Rightarrow \frac{1}{4}C+D=4$

Let $x=-\frac{1}{4}: \frac{7}{4} = \left(-\frac{1}{4}C+D\right)\times\frac{7}{16} \Rightarrow -\frac{1}{4}C+D=4$

$A=0,\ B=-3,\ C=0,\ D=4$

$\therefore \frac{12x^2+1}{(1-9x^2)(1-16x^2)} \equiv -\frac{3}{1-9x^2}+\frac{4}{1-16x^2}$

$\int \frac{12x^2+1}{(1-9x^2)(1-16x^2)}\,\mathrm{d}x = \int \left(-\frac{3}{1-9x^2}+\frac{4}{1-16x^2}\right)\mathrm{d}x$

$= \int \left(-\frac{3}{1-(3x)^2}+\frac{4}{1-(4x)^2}\right)\mathrm{d}x$

$= -3\times\frac{1}{3}\operatorname{artanh} 3x + 4\times\frac{1}{4}\times\operatorname{artanh} 4x + c$

$= -\operatorname{artanh} 3x + \operatorname{artanh} 4x + c$

14) b)

$\frac{10x^2+47}{(1-4x^2)(25-x^2)} \equiv \frac{Ax+B}{1-4x^2} + \frac{Cx+D}{25-x^2}$

$\Rightarrow 10x^2 + 47 \equiv (Ax + B)(25 - x^2) + (Cx + D)(1 - 4x^2)$

Let $x = 5$: $297 = (5C + D) \times (-99) \Rightarrow 5C + D = -3$

Let $x = -5$: $297 = (-5C + D) \times (-99) \Rightarrow -5C + D = -3$

Let $x = \frac{1}{2}$: $\frac{99}{2} = \left(\frac{1}{2}A + B\right) \times \frac{99}{4} \Rightarrow \frac{1}{2}A + B = 2$

Let $x = -\frac{1}{2}$: $\frac{99}{2} = \left(-\frac{1}{2}A + B\right) \times \frac{99}{4} \Rightarrow -\frac{1}{2}A + B = 2$

$A = 0, B = 2, C = 0, D = -3$

$\therefore \frac{10x^2+47}{(1-4x^2)(25-x^2)} \equiv \frac{2}{1-4x^2} - \frac{3}{25-x^2}$

$\int \frac{10x^2+47}{(1-4x^2)(25-x^2)} \mathrm{d}x = \int \left(\frac{2}{1-4x^2} - \frac{3}{25-x^2}\right) \mathrm{d}x$

$= \int \left(\frac{2}{1-(2x)^2} - \frac{3}{25\left(1-\frac{1}{25}x^2\right)}\right) \mathrm{d}x$

$= \int \left(\frac{2}{1-(2x)^2} - \frac{3}{25} \times \frac{1}{1-\left(\frac{1}{5}x\right)^2}\right) \mathrm{d}x$

$= 2 \times \frac{1}{2} \operatorname{artanh} 2x - \frac{3}{25} \times \frac{1}{\frac{1}{5}} \times \operatorname{artanh}\left(\frac{1}{5}x\right) + c$

$= \operatorname{artanh} 2x - \frac{3}{5} \operatorname{artanh}\left(\frac{1}{5}x\right) + c$

14) c)

$\frac{108x^4+12x^3-237x^2-27x+25}{(1-9x^2)(9-4x^2)} \equiv E + \frac{Ax+B}{1-9x^2} + \frac{Cx+D}{9-4x^2}$

The coefficient of x^4 in the numerator is 108 and the coefficient of x^4 in the denominator is 36, so $E = \frac{108}{36} = 3$

$\therefore \frac{108x^4+12x^3-237x^2-27x+25}{(1-9x^2)(9-4x^2)} \equiv 3 + \frac{Ax+B}{1-9x^2} + \frac{Cx+D}{9-4x^2}$

$\Rightarrow 108x^4 + 12x^3 - 237x^2 - 27x + 25 \equiv 3(1-9x^2)(9-4x^2) + (Ax+B)(9-4x^2) + (Cx+D)(1-9x^2)$

Let $x = \frac{1}{3}$: $-\frac{77}{9} = 0 + \left(\frac{1}{3}A + B\right) \times \frac{77}{9} \Rightarrow \frac{1}{3}A + B = -1$

Let $x = -\frac{1}{3}$: $\frac{77}{9} = 0 + \left(-\frac{1}{3}A + B\right) \times \frac{77}{9} \Rightarrow -\frac{1}{3}A + B = 1$

Let $x = \frac{3}{2}$: $\frac{77}{2} = 0 + \left(\frac{3}{2}C + D\right) \times \left(-\frac{77}{4}\right) \Rightarrow \frac{3}{2}C + D = -2$

Let $x = -\frac{3}{2}$: $\frac{77}{2} = 0 + \left(-\frac{3}{2}C + D\right) \times \left(-\frac{77}{4}\right) \Rightarrow -\frac{3}{2}C + D = -2$

$A = -3$, $B = 0$, $C = 0$, $D = -2$

$\therefore \frac{108x^4+12x^3-237x^2-27x+25}{(1-9x^2)(9-4x^2)} \equiv 3 - \frac{3x}{1-9x^2} - \frac{2}{9-4x^2}$

$\int \frac{108x^4+12x^3-237x^2-27x+25}{(1-9x^2)(9-4x^2)} \mathrm{d}x = \int \left(3 - \frac{3x}{1-9x^2} - \frac{2}{9-4x^2}\right) \mathrm{d}x$

$= \int \left(3 - \frac{3x}{1-9x^2} - \frac{2}{9\left(1-\frac{4}{9}x^2\right)}\right) \mathrm{d}x$

$= \int \left(3 - \frac{3x}{1-9x^2} - \frac{2}{9} \times \frac{1}{1-\left(\frac{2}{3}x\right)^2}\right) \mathrm{d}x$

$= 3x + \frac{1}{6}\ln|1-9x^2| - \frac{2}{9} \times \frac{1}{\frac{2}{3}} \times \operatorname{artanh}\left(\frac{2}{3}x\right) + c$

$= 3x + \frac{1}{6}\ln|1-9x^2| - \frac{1}{3}\operatorname{artanh}\left(\frac{2}{3}x\right) + c$

15)

$\int \frac{1-2x}{1-x^2} \mathrm{d}x$

$= \int \left(\frac{1}{1-x^2} - \frac{2x}{1-x^2}\right) \mathrm{d}x$

$= \operatorname{artanh} x + \ln|1-x^2| + c$

16) a)

$$\int \frac{6x+1}{1-9x^2}\,\mathrm{d}x$$
$$= \int \left(\frac{6x}{1-9x^2} + \frac{1}{1-9x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{6x}{1-9x^2} + \frac{1}{1-(3x)^2}\right)\mathrm{d}x$$
$$= -\frac{1}{3}\ln|1-9x^2| + \frac{1}{3}\operatorname{artanh} 3x + c$$

16) b)

$$\int \frac{2-8x}{4-x^2}\,\mathrm{d}x$$
$$= \int \left(\frac{2}{4-x^2} - \frac{8x}{4-x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{2}{4\left(1-\frac{1}{4}x^2\right)} - \frac{8x}{4-x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{1}{2} \times \frac{1}{1-\left(\frac{1}{2}x\right)^2} - \frac{8x}{4-x^2}\right)\mathrm{d}x$$
$$= \frac{1}{2} \times \frac{1}{\frac{1}{2}} \times \operatorname{artanh}\left(\frac{1}{2}x\right) + 4\ln|4-x^2| + c$$
$$= \operatorname{artanh}\left(\frac{1}{2}x\right) + 4\ln|4-x^2| + c$$

16) c)

$$\int \frac{8-3x}{16-9x^2}\,\mathrm{d}x$$
$$= \int \left(\frac{8}{16-9x^2} - \frac{3x}{16-9x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{8}{16\left(1-\frac{9}{16}x^2\right)} - \frac{3x}{16-9x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{1}{2} \times \frac{1}{1-\left(\frac{3}{4}x\right)^2} - \frac{3x}{16-9x^2}\right)\mathrm{d}x$$
$$= \frac{1}{2} \times \frac{1}{\frac{3}{4}} \times \operatorname{artanh}\left(\frac{3}{4}x\right) + \frac{1}{6}\ln|16-9x^2| + c$$
$$= \frac{2}{3}\operatorname{artanh}\left(\frac{3}{4}x\right) + \frac{1}{6}\ln|16-9x^2| + c$$

16) d)

$$\int \frac{10x-4}{25-100x^2}\,\mathrm{d}x$$
$$= \int \left(\frac{10x}{25-100x^2} - \frac{4}{25-100x^2}\right)\mathrm{d}x$$
$$= \int \left(\frac{10x}{25-100x^2} - \frac{4}{25(1-4x^2)}\right)\mathrm{d}x$$
$$= \int \left(\frac{10x}{25-100x^2} - \frac{4}{25}\times\frac{1}{1-(2x)^2}\right)\mathrm{d}x$$
$$= -\frac{1}{20}\ln|25-100x^2| - \frac{4}{25}\times\frac{1}{2}\times \operatorname{artanh} 2x + c$$
$$= -\frac{1}{20}\ln|25-100x^2| - \frac{2}{25}\operatorname{artanh} 2x + c$$

17)

$$\int \frac{1+x}{\sqrt{x^2-1}}\,\mathrm{d}x$$
$$= \int \left(\frac{1}{\sqrt{x^2-1}} + \frac{x}{\sqrt{x^2-1}}\right)\mathrm{d}x$$
$$= \operatorname{arcosh} x + \sqrt{x^2-1} + c$$

18)

$$\int \frac{4x+1}{\sqrt{x^2+1}}\,\mathrm{d}x$$
$$= \int \left(\frac{4x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}}\right)\mathrm{d}x$$
$$= 4\sqrt{x^2+1} + \operatorname{arsinh} x + c$$

19) a)

$$\int \frac{2x-1}{\sqrt{4x^2-1}}\,\mathrm{d}x$$
$$= \int \left(\frac{2x}{\sqrt{4x^2-1}} - \frac{1}{\sqrt{4x^2-1}}\right)\mathrm{d}x$$
$$= \int \left(\frac{2x}{\sqrt{4x^2-1}} - \frac{1}{\sqrt{(2x)^2-1}}\right)\mathrm{d}x$$
$$= \frac{1}{2}\sqrt{4x^2-1} - \frac{1}{2}\operatorname{arcosh} 2x + c$$

19) b)

$$\int \frac{1-3x}{\sqrt{9x^2+1}}\,\mathrm{d}x$$
$$= \int \left(\frac{1}{\sqrt{9x^2+1}} - \frac{3x}{\sqrt{9x^2+1}}\right)\mathrm{d}x$$
$$= \int \left(\frac{1}{\sqrt{(3x)^2+1}} - \frac{3x}{\sqrt{9x^2+1}}\right)\mathrm{d}x$$
$$= \frac{1}{3}\operatorname{arsinh} 3x - \frac{1}{3}\sqrt{9x^2+1} + c$$

19) c)

$$\int \frac{4+6x}{\sqrt{9x^2-4}}\,\mathrm{d}x$$
$$= \int \left(\frac{4}{\sqrt{9x^2-4}} + \frac{6x}{\sqrt{9x^2-4}}\right)\mathrm{d}x$$
$$= \int \left(\frac{4}{\sqrt{4\left(\frac{9}{4}x^2-1\right)}} + \frac{6x}{\sqrt{9x^2-4}}\right)\mathrm{d}x$$
$$= \int \left(\frac{2}{\sqrt{\left(\frac{3}{2}x\right)^2-1}} + \frac{6x}{\sqrt{9x^2-4}}\right)\mathrm{d}x$$
$$= 2 \times \frac{1}{\frac{3}{2}} \times \operatorname{arcosh}\left(\frac{3}{2}x\right) + \frac{2}{3}\sqrt{9x^2-4} + c$$
$$= \frac{4}{3}\operatorname{arcosh}\left(\frac{3}{2}x\right) + \frac{2}{3}\sqrt{9x^2-4} + c$$

19) d)

$$\int \frac{8x-3}{\sqrt{4x^2+36}}\,\mathrm{d}x$$
$$= \int \left(\frac{8x}{\sqrt{4x^2+36}} - \frac{3}{\sqrt{4x^2+36}}\right)\mathrm{d}x$$
$$= \int \left(\frac{8x}{\sqrt{4x^2+36}} - \frac{3}{\sqrt{36\left(\frac{1}{9}x^2+1\right)}}\right)\mathrm{d}x$$
$$= \int \left(\frac{8x}{\sqrt{4x^2+36}} - \frac{1}{2} \times \frac{1}{\sqrt{\left(\frac{1}{3}x\right)^2+1}}\right)\mathrm{d}x$$
$$= 2\sqrt{4x^2+36} - \frac{1}{2} \times \frac{1}{\frac{1}{3}} \times \operatorname{arsinh}\left(\frac{1}{3}x\right) + c$$
$$= 2\sqrt{4x^2+36} - \frac{3}{2}\operatorname{arsinh}\left(\frac{1}{3}x\right) + c$$

20) a)

Let $x = \sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u \, \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{x^2+1}} \mathrm{d}x$

$= \int \frac{1}{\sqrt{\sinh^2 u+1}} \times \cosh u \, \mathrm{d}u$

$= \int \frac{\cosh u}{\sqrt{\cosh^2 u}} \mathrm{d}u$

$= \int \frac{\cosh u}{\cosh u} \mathrm{d}u$

$= \int 1 \, \mathrm{d}u$

$= u + c$

$= \operatorname{arsinh} x + c$

So $x = \sinh u$ gives $\int \frac{1}{\sqrt{x^2+1}} \mathrm{d}x = \operatorname{arsinh} x + c$

20) b)

Let $x = \cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sinh u \Rightarrow \mathrm{d}x = \sinh u \, \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{x^2-1}} \mathrm{d}x$

$= \int \frac{1}{\sqrt{\cosh^2 u-1}} \times \sinh u \, \mathrm{d}u$

$= \int \frac{\sinh u}{\sqrt{\sinh^2 u}} \mathrm{d}u$

$= \int \frac{\sinh u}{\sinh u} \mathrm{d}u$

$= \int 1 \, \mathrm{d}u$

$= u + c$

$= \operatorname{arcosh} x + c$

So $x = \cosh u$ gives $\int \frac{1}{\sqrt{x^2-1}} \mathrm{d}x = \operatorname{arcosh} x + c$

20) c)

Let $x = \tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \operatorname{sech}^2 u \, \mathrm{d}u$

$\therefore \int \frac{1}{1-x^2} \mathrm{d}x$

$= \int \frac{1}{1-\tanh^2 u} \times \operatorname{sech}^2 u \, \mathrm{d}u$

$= \int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u} \mathrm{d}u$

$= \int 1 \, \mathrm{d}u$

$= u + c$

$= \operatorname{artanh} x + c$

So $x = \tanh u$ gives $\int \frac{1}{1-x^2} \mathrm{d}x = \operatorname{artanh} x + c$

20) d)

Let $4x^2 = \sinh^2 u \Rightarrow 2x = \sinh u \Rightarrow x = \frac{1}{2}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{2}\cosh u \Rightarrow \mathrm{d}x = \frac{1}{2}\cosh u \, \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{4x^2+1}} \mathrm{d}x$

$= \int \frac{1}{\sqrt{\sinh^2 u+1}} \times \frac{1}{2}\cosh u \, \mathrm{d}u$

$= \frac{1}{2}\int \frac{\cosh u}{\sqrt{\cosh^2 u}} \mathrm{d}u$

$= \frac{1}{2}\int \frac{\cosh u}{\cosh u} \mathrm{d}u$

$= \int \frac{1}{2} \mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\operatorname{arsinh} 2x + c$

So $x = \frac{1}{2}\sinh u$ gives $\int \frac{1}{\sqrt{4x^2+1}} \mathrm{d}x = \frac{1}{2}\operatorname{arsinh} 2x + c$

20) e)

Let $x^2 = 9\cosh^2 u \Rightarrow x = 3\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 3\sinh u \Rightarrow \mathrm{d}x = 3\sinh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{x^2-9}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9\cosh^2 u-9}} \times 3\sinh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{9(\cosh^2 u-1)}} \times 3\sinh u\,\mathrm{d}u$

$= \int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int 1\,\mathrm{d}u$

$= u + c$

$= \operatorname{arcosh}\left(\frac{1}{3}x\right) + c$

So $x = 3\cosh u$ gives $\int \frac{1}{\sqrt{x^2-9}}\mathrm{d}x = \operatorname{arcosh}\left(\frac{1}{3}x\right) + c$

20) f)

Let $9x^2 = \tanh^2 u \Rightarrow 3x = \tanh u \Rightarrow x = \frac{1}{3}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{3}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{1}{3}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{1-9x^2}\mathrm{d}x$

$= \int \frac{1}{1-\tanh^2 u} \times \frac{1}{3}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{1}{3}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int \frac{1}{3}\mathrm{d}u$

$= \frac{1}{3}u + c$

$= \frac{1}{3}\operatorname{artanh} 3x + c$

So $x = \frac{1}{3}\tanh u$ gives $\int \frac{1}{1-9x^2}\mathrm{d}x = \frac{1}{3}\operatorname{artanh} 3x + c$

20) g)

Let $x^2 = 25\sinh^2 u \Rightarrow x = 5\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 5\cosh u \Rightarrow \mathrm{d}x = 5\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{x^2+25}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{25\sinh^2 u+25}} \times 5\cosh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{25(\sinh^2 u+1)}} \times 5\cosh u\,\mathrm{d}u$

$= \int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int 1\,\mathrm{d}u$

$= u + c$

$= \operatorname{arsinh}\left(\frac{1}{5}x\right) + c$

So $x = 5\sinh u$ gives $\int \frac{1}{\sqrt{x^2+25}}\mathrm{d}x = \operatorname{arsinh}\left(\frac{1}{5}x\right) + c$

20) h)

Let $36x^2 = \cosh^2 u \Rightarrow 6x = \cosh u \Rightarrow x = \frac{1}{6}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{6}\sinh u \Rightarrow \mathrm{d}x = \frac{1}{6}\sinh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{36x^2-9}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{\cosh^2 u-1}} \times \frac{1}{6}\sinh u\,\mathrm{d}u$

$= \frac{1}{6}\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \frac{1}{6}\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int \frac{1}{6}\mathrm{d}u$

$= \frac{1}{6}u + c$

$= \frac{1}{6}\operatorname{arcosh} 6x + c$

So $x = \frac{1}{6}\cosh u$ gives $\int \frac{1}{\sqrt{36x^2-9}}\mathrm{d}x = \frac{1}{6}\operatorname{arcosh} 6x + c$

20) i)

Let $x^2 = 100\tanh^2 u \Rightarrow x = 10\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 10\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = 10\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{100-x^2}\mathrm{d}x$

$= \int \frac{1}{100-100\tanh^2 u} \times 10\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{1}{100(1-\tanh^2 u)} \times 10\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{1}{10}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int \frac{1}{10}\mathrm{d}u$

$= \frac{1}{10}u + c$

$= \frac{1}{10}\operatorname{artanh}\left(\frac{1}{10}x\right) + c$

So $x = 10\tanh u$ gives $\int \frac{1}{100-x^2}\mathrm{d}x = \frac{1}{10}\operatorname{artanh}\left(\frac{1}{10}x\right) + c$

20) j)

Let $49x^2 = \sinh^2 u \Rightarrow 7x = \sinh u \Rightarrow x = \frac{1}{7}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{7}\cosh u \Rightarrow \mathrm{d}x = \frac{1}{7}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{49x^2+1}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{\sinh^2 u+1}} \times \frac{1}{7}\cosh u\,\mathrm{d}u$

$= \frac{1}{7}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{1}{7}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int \frac{1}{7}\mathrm{d}u$

$= \frac{1}{7}u + c$

$= \frac{1}{7}\operatorname{arsinh} 7x + c$

So $x = \frac{1}{7}\sinh u$ gives $\int \frac{1}{\sqrt{49x^2+1}}\mathrm{d}x = \frac{1}{7}\operatorname{arsinh} 7x + c$

20) k)

Let $x^2 = 64\cosh^2 u \Rightarrow x = 8\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 8\sinh u \Rightarrow \mathrm{d}x = 8\sinh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{x^2-64}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{64\cosh^2 u-64}} \times 8\sinh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{64(\cosh^2 u-1)}} \times 8\sinh u\,\mathrm{d}u$

$= \int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int 1\,\mathrm{d}u$

$= u + c$

$= \operatorname{arcosh}\left(\frac{1}{8}x\right) + c$

So $x = 8\cosh u$ gives $\int \frac{1}{\sqrt{x^2-64}}\mathrm{d}x = \operatorname{arcosh}\left(\frac{1}{8}x\right) + c$

20) l)

Let $144x^2 = \tanh^2 u \Rightarrow 12x = \tanh u \Rightarrow x = \frac{1}{12}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{12}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{1}{12}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{1-144x^2}\mathrm{d}x$

$= \int \frac{1}{1-\tanh^2 u} \times \frac{1}{12}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{1}{12}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int \frac{1}{12}\mathrm{d}u$

$= \frac{1}{12}u + c$

$= \frac{1}{12}\operatorname{artanh} 12x + c$

So $x = \frac{1}{12}\tanh u$ gives $\int \frac{1}{1-144x^2}\mathrm{d}x = \frac{1}{12}\operatorname{artanh} 12x + c$

20) m)

Let $4x^2 = 9\sinh^2 u \Rightarrow 2x = 3\sinh u \Rightarrow x = \frac{3}{2}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{2}\cosh u \Rightarrow \mathrm{d}x = \frac{3}{2}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{4x^2+9}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9\sinh^2 u+9}} \times \frac{3}{2}\cosh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{9(\sinh^2 u+1)}} \times \frac{3}{2}\cosh u\,\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int \frac{1}{2}\mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\operatorname{arsinh}\left(\frac{2}{3}x\right) + c$

So $x = \frac{3}{2}\sinh u$ gives $\int \frac{1}{\sqrt{4x^2+9}}\mathrm{d}x = \frac{1}{2}\operatorname{arsinh}\left(\frac{2}{3}x\right) + c$

20) n)

Let $9x^2 = 16\cosh^2 u \Rightarrow 3x = 4\cosh u \Rightarrow x = \frac{4}{3}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{4}{3}\sinh u \Rightarrow \mathrm{d}x = \frac{4}{3}\sinh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{9x^2-16}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{16\cosh^2 u-16}} \times \frac{4}{3}\sinh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{16(\cosh^2 u-1)}} \times \frac{4}{3}\sinh u\,\mathrm{d}u$

$= \frac{1}{3}\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \frac{1}{3}\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int \frac{1}{3}\mathrm{d}u$

$= \frac{1}{3}u + c$

$= \frac{1}{3}\operatorname{arcosh}\left(\frac{3}{4}x\right) + c$

So $x = \frac{4}{3}\cosh u$ gives $\int \frac{1}{\sqrt{9x^2-16}}\mathrm{d}x = \frac{1}{3}\operatorname{arcosh}\left(\frac{3}{4}x\right) + c$

20) o)

Let $25x^2 = 4\tanh^2 u \Rightarrow 5x = 2\tanh u \Rightarrow x = \frac{2}{5}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2}{5}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{2}{5}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{4-25x^2}\mathrm{d}x$

$= \int \frac{1}{4-4\tanh^2 u} \times \frac{2}{5}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{1}{4(1-\tanh^2 u)} \times \frac{2}{5}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{1}{10}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int \frac{1}{10}\mathrm{d}u$

$= \frac{1}{10}u + c$

$= \frac{1}{10}\operatorname{artanh}\left(\frac{5}{2}x\right) + c$

So $x = \frac{2}{5}\tanh u$ gives $\int \frac{1}{4-25x^2}\mathrm{d}x = \frac{1}{10}\operatorname{artanh}\left(\frac{5}{2}x\right) + c$

20) p)

Let $36x^2 = 25\sinh^2 u \Rightarrow 6x = 5\sinh u \Rightarrow x = \frac{5}{6}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{5}{6}\cosh u \Rightarrow \mathrm{d}x = \frac{5}{6}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{36x^2+25}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{25\sinh^2 u+25}} \times \frac{5}{6}\cosh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{25(\sinh^2 u+1)}} \times \frac{5}{6}\cosh u\,\mathrm{d}u$

$= \frac{1}{6}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{1}{6}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int \frac{1}{6}\mathrm{d}u$

$= \frac{1}{6}u + c$

$= \frac{1}{6}\operatorname{arsinh}\left(\frac{6}{5}x\right) + c$

So $x = \frac{5}{6}\sinh u$ gives $\int \frac{1}{\sqrt{36x^2+25}}\mathrm{d}x = \frac{1}{6}\operatorname{arsinh}\left(\frac{6}{5}x\right) + c$

20) q)

Let $400x^2 = 9\cosh^2 u \Rightarrow 20x = 3\cosh u \Rightarrow x = \frac{3}{20}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{20}\sinh u \Rightarrow \mathrm{d}x = \frac{3}{20}\sinh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{400x^2-9}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9\cosh^2 u-9}} \times \frac{3}{20}\sinh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{9(\cosh^2 u-1)}} \times \frac{3}{20}\sinh u\,\mathrm{d}u$

$= \frac{1}{20}\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \frac{1}{20}\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int \frac{1}{20}\mathrm{d}u$

$= \frac{1}{20}u + c$

$= \frac{1}{20}\operatorname{arcosh}\left(\frac{20}{3}x\right) + c$

So $x = \frac{3}{20}\cosh u$ gives $\int \frac{1}{\sqrt{400x^2-9}}\mathrm{d}x = \frac{1}{20}\operatorname{arcosh}\left(\frac{20}{3}x\right) + c$

20) r)

Let $36x^2 = 81\tanh^2 u \Rightarrow 6x = 9\tanh u \Rightarrow x = \frac{3}{2}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{2}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{3}{2}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{81-36x^2}\mathrm{d}x$

$= \int \frac{1}{81-81\tanh^2 u} \times \frac{3}{2}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{1}{81(1-\tanh^2 u)} \times \frac{3}{2}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{1}{54}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int \frac{1}{54}\mathrm{d}u$

$= \frac{1}{54}u + c$

$= \frac{1}{54}\operatorname{artanh}\left(\frac{2}{3}x\right) + c$

So $x = \frac{3}{2}\tanh u$ gives $\int \frac{1}{81-36x^2}\mathrm{d}x = \frac{1}{54}\operatorname{artanh}\left(\frac{2}{3}x\right) + c$

20) s)

Let $2x^2 = 9\sinh^2 u \Rightarrow \sqrt{2}x = 3\sinh u \Rightarrow x = \frac{3}{\sqrt{2}}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3\sqrt{2}}{2}\cosh u \Rightarrow \mathrm{d}x = \frac{3\sqrt{2}}{2}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{2x^2+9}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9\sinh^2 u+9}} \times \frac{3\sqrt{2}}{2}\cosh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{9(\sinh^2 u+1)}} \times \frac{3\sqrt{2}}{2}\cosh u\,\mathrm{d}u$

$= \frac{\sqrt{2}}{2}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{\sqrt{2}}{2}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int \frac{\sqrt{2}}{2}\mathrm{d}u$

$= \frac{\sqrt{2}}{2}u + c$

$= \frac{\sqrt{2}}{2}\operatorname{arsinh}\left(\frac{\sqrt{2}}{3}x\right) + c$

So $x = \frac{3}{\sqrt{2}}\sinh u$ gives $\int \frac{1}{\sqrt{2x^2+9}}\mathrm{d}x = \frac{\sqrt{2}}{2}\operatorname{arsinh}\left(\frac{\sqrt{2}}{3}x\right) + c$

20) t)

Let $25x^2 = 3\cosh^2 u \Rightarrow 5x = \sqrt{3}\cosh u \Rightarrow x = \frac{\sqrt{3}}{5}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{3}}{5}\sinh u \Rightarrow \mathrm{d}x = \frac{\sqrt{3}}{5}\sinh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{25x^2-3}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{3\cosh^2 u-3}} \times \frac{\sqrt{3}}{5}\sinh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{3(\cosh^2 u-1)}} \times \frac{\sqrt{3}}{5}\sinh u\,\mathrm{d}u$

$= \frac{1}{5}\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \frac{1}{5}\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int \frac{1}{5}\mathrm{d}u$

$= \frac{1}{5}u + c$

$= \frac{1}{5}\operatorname{arcosh}\left(\frac{5}{\sqrt{3}}x\right) + c$

$= \frac{1}{5}\operatorname{arcosh}\left(\frac{5\sqrt{3}}{3}x\right) + c$

So $x = \frac{\sqrt{3}}{5}\cosh u$ gives $\int \frac{1}{\sqrt{25x^2-3}}\mathrm{d}x = \frac{1}{5}\operatorname{arcosh}\left(\frac{5\sqrt{3}}{3}x\right) + c$

20) u)

Let $5x^2 = 9\tanh^2 u \Rightarrow \sqrt{5}x = 3\tanh u \Rightarrow x = \frac{3}{\sqrt{5}}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3\sqrt{5}}{5}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{3\sqrt{5}}{5}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{9-5x^2}\mathrm{d}x$

$= \int \frac{1}{9-9\tanh^2 u} \times \frac{3\sqrt{5}}{5}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{1}{9(1-\tanh^2 u)} \times \frac{3\sqrt{5}}{5}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{\sqrt{5}}{15}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int \frac{\sqrt{5}}{15}\mathrm{d}u$

$= \frac{\sqrt{5}}{15}u + c$

$= \frac{\sqrt{5}}{15}\operatorname{artanh}\left(\frac{\sqrt{5}}{3}x\right) + c$

So $x = \frac{3}{\sqrt{5}}\tanh u$ gives $\int \frac{1}{9-5x^2}\mathrm{d}x = \frac{\sqrt{5}}{15}\operatorname{artanh}\left(\frac{\sqrt{5}}{3}x\right) + c$

20) v)

Let $3x^2 = 8\sinh^2 u \Rightarrow \sqrt{3}x = \sqrt{8}\sinh u \Rightarrow x = \frac{\sqrt{8}}{\sqrt{3}}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2\sqrt{6}}{3}\cosh u \Rightarrow \mathrm{d}x = \frac{2\sqrt{6}}{3}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{3x^2+8}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{8\sinh^2 u+8}} \times \frac{2\sqrt{6}}{3}\cosh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{8(\sinh^2 u+1)}} \times \frac{2\sqrt{6}}{3}\cosh u\,\mathrm{d}u$

$= \frac{\sqrt{3}}{3}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{\sqrt{3}}{3}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int \frac{\sqrt{3}}{3}\mathrm{d}u$

$= \frac{\sqrt{3}}{3}u + c$

$= \frac{\sqrt{3}}{3}\operatorname{arsinh}\left(\frac{\sqrt{3}}{\sqrt{8}}x\right) + c$

$= \frac{\sqrt{3}}{3}\operatorname{arsinh}\left(\frac{\sqrt{6}}{4}x\right) + c$

So $x = \frac{\sqrt{8}}{\sqrt{3}}\sinh u$ gives $\int \frac{1}{\sqrt{3x^2+8}}\mathrm{d}x = \frac{\sqrt{3}}{3}\operatorname{arsinh}\left(\frac{\sqrt{6}}{4}x\right) + c$

20) w)

Let $12x^2 = 15\cosh^2 u \Rightarrow \sqrt{12}x = \sqrt{15}\cosh u \Rightarrow x = \frac{\sqrt{15}}{\sqrt{12}}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{5}}{2}\sinh u \Rightarrow \mathrm{d}x = \frac{\sqrt{5}}{2}\sinh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{12x^2-15}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{15\cosh^2 u-15}} \times \frac{\sqrt{5}}{2}\sinh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{15(\cosh^2 u-1)}} \times \frac{\sqrt{5}}{2}\sinh u\,\mathrm{d}u$

$= \frac{\sqrt{3}}{6}\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \frac{\sqrt{3}}{6}\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int \frac{\sqrt{3}}{6}\mathrm{d}u$

$= \frac{\sqrt{3}}{6}u + c$

$= \frac{\sqrt{3}}{6}\operatorname{arcosh}\left(\frac{\sqrt{12}}{\sqrt{15}}x\right) + c$

$= \frac{\sqrt{3}}{6}\operatorname{arcosh}\left(\frac{2\sqrt{5}}{5}x\right) + c$

So $x = \frac{\sqrt{15}}{\sqrt{12}}\cosh u$ gives $\int \frac{1}{\sqrt{12x^2-15}}\mathrm{d}x = \frac{\sqrt{3}}{6}\operatorname{arcosh}\left(\frac{2\sqrt{5}}{5}x\right) + c$

20) x)

Let $7x^2 = 24\tanh^2 u \Rightarrow \sqrt{7}x = \sqrt{24}\tanh u \Rightarrow x = \frac{\sqrt{24}}{\sqrt{7}}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2\sqrt{42}}{7}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{2\sqrt{42}}{7}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{24-7x^2}\mathrm{d}x$

$= \int \frac{1}{24-24\tanh^2 u} \times \frac{2\sqrt{42}}{7}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{1}{24(1-\tanh^2 u)} \times \frac{2\sqrt{42}}{7}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{\sqrt{42}}{84}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int \frac{\sqrt{42}}{84}\mathrm{d}u$

$= \frac{\sqrt{42}}{84}u + c$

$= \frac{\sqrt{42}}{84}\operatorname{artanh}\left(\frac{\sqrt{7}}{\sqrt{24}}x\right) + c$

$= \frac{\sqrt{42}}{84}\operatorname{artanh}\left(\frac{\sqrt{42}}{12}x\right) + c$

So $x = \frac{\sqrt{24}}{\sqrt{7}}\tanh u$ gives $\int \frac{1}{24-7x^2}\mathrm{d}x = \frac{\sqrt{42}}{84}\operatorname{artanh}\left(\frac{\sqrt{42}}{12}x\right) + c$

20) y)

Let $bx^2 = a\tanh^2 u \Rightarrow \sqrt{b}x = \sqrt{a}\tanh u \Rightarrow x = \frac{\sqrt{a}}{\sqrt{b}}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{a}}{\sqrt{b}}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{\sqrt{a}}{\sqrt{b}}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{a-7x^2}\mathrm{d}x$

$= \int \frac{1}{a-a\tanh^2 u} \times \frac{\sqrt{a}}{\sqrt{b}}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{1}{a(1-\tanh^2 u)} \times \frac{\sqrt{a}}{\sqrt{b}}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{\sqrt{a}}{a\sqrt{b}}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \int \frac{\sqrt{a}}{a\sqrt{b}}\mathrm{d}u$

$= \frac{\sqrt{a}}{a\sqrt{b}}u + c$

$= \frac{\sqrt{a}}{a\sqrt{b}}\operatorname{artanh}\left(\frac{\sqrt{a}}{\sqrt{b}}x\right) + c$

$= \frac{1}{\sqrt{ab}}\operatorname{artanh}\left(\sqrt{\frac{a}{b}}x\right) + c$

So $x = \frac{\sqrt{a}}{\sqrt{b}}\tanh u$ gives $\int \frac{1}{a-bx^2}\mathrm{d}x = \frac{1}{\sqrt{ab}}\operatorname{artanh}\left(\sqrt{\frac{a}{b}}x\right) + c$

20) z)

Let $ax^2 = b\sinh^2 u \Rightarrow \sqrt{a}x = \sqrt{b}\sinh u \Rightarrow x = \frac{\sqrt{b}}{\sqrt{a}}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{b}}{\sqrt{a}}\cosh u \Rightarrow \mathrm{d}x = \frac{\sqrt{b}}{\sqrt{a}}\cosh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{ax^2+b}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{b\sinh^2 u+b}} \times \frac{\sqrt{b}}{\sqrt{a}}\cosh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{b(\sinh^2 u+1)}} \times \frac{\sqrt{b}}{\sqrt{a}}\cosh u\,\mathrm{d}u$

$= \frac{\sqrt{b}}{\sqrt{ab}}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{\sqrt{b}}{\sqrt{ab}}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \int \frac{\sqrt{b}}{\sqrt{ab}}\mathrm{d}u$

$= \frac{\sqrt{b}}{\sqrt{ab}}u + c$

$= \frac{\sqrt{b}}{\sqrt{ab}}\operatorname{arsinh}\left(\frac{\sqrt{a}}{\sqrt{b}}x\right) + c$

$= \frac{1}{\sqrt{a}}\operatorname{arsinh}\left(\sqrt{\frac{a}{b}}x\right) + c$

So $x = \frac{\sqrt{b}}{\sqrt{a}}\sinh u$ gives $\int \frac{1}{\sqrt{ax^2+b}}\mathrm{d}x = \frac{1}{\sqrt{a}}\operatorname{arsinh}\left(\sqrt{\frac{a}{b}}x\right) + c$

20) aa)

Let $ax^2 = b\cosh^2 u \Rightarrow \sqrt{a}x = \sqrt{b}\cosh u \Rightarrow x = \frac{\sqrt{b}}{\sqrt{a}}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{\sqrt{b}}{\sqrt{a}}\sinh u \Rightarrow \mathrm{d}x = \frac{\sqrt{b}}{\sqrt{a}}\sinh u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{ax^2-b}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{b\cosh^2 u - b}} \times \frac{\sqrt{b}}{\sqrt{a}}\sinh u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{b(\cosh^2 u - 1)}} \times \frac{\sqrt{b}}{\sqrt{a}}\sinh u\,\mathrm{d}u$

$= \frac{\sqrt{b}}{\sqrt{ab}}\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \frac{\sqrt{b}}{\sqrt{ab}}\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \int \frac{\sqrt{b}}{\sqrt{ab}}\mathrm{d}u$

$= \frac{\sqrt{b}}{\sqrt{ab}}u + c$

$= \frac{\sqrt{b}}{\sqrt{ab}}\operatorname{arcosh}\left(\frac{\sqrt{a}}{\sqrt{b}}x\right) + c$

$= \frac{1}{\sqrt{a}}\operatorname{arcosh}\left(\sqrt{\frac{a}{b}}x\right) + c$

So $x = \frac{\sqrt{b}}{\sqrt{a}}\cosh u$ gives $\int \frac{1}{\sqrt{ax^2-b}}\mathrm{d}x = \frac{1}{\sqrt{a}}\operatorname{arcosh}\left(\sqrt{\frac{a}{b}}x\right) + c$

21) a)

Let $x^2 = \sinh^2 u \Rightarrow x = \sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u \, \mathrm{d}u$

$\therefore I = \int \sqrt{x^2 + 1} \, \mathrm{d}x$

$= \int \sqrt{\sinh^2 u + 1} \times \cosh u \, \mathrm{d}u$

$= \int \sqrt{\cosh^2 u} \times \cosh u \, \mathrm{d}u$

$= \int \cosh^2 u \, \mathrm{d}u$

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A \equiv \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore I = \int \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right) \mathrm{d}u$

$= \frac{1}{4}\sinh 2u + \frac{1}{2}u + c$

$= \frac{1}{4}(2\sinh u \cosh u) + \frac{1}{2}u + c$

$= \frac{1}{2}\sinh u \cosh u + \frac{1}{2}u + c$

$= \frac{1}{2}\sinh u \sqrt{\sinh^2 u + 1} + \frac{1}{2}u + c$

$= \frac{1}{2}x\sqrt{x^2 + 1} + \frac{1}{2}\operatorname{arsinh} x + c$

So $x = \sinh u$ gives $\int \sqrt{x^2 + 1} \, \mathrm{d}x = \frac{1}{2}x\sqrt{x^2 + 1} + \frac{1}{2}\operatorname{arsinh} x + c$

21) b)

Let $x^2 = \sinh^2 u \Rightarrow x = \sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u \, \mathrm{d}u$

$\therefore I = \int (x^2 + 1)^{\frac{3}{2}} \mathrm{d}x$

$= \int (\sinh^2 u + 1)^{\frac{3}{2}} \times \cosh u \, \mathrm{d}u$

$= \int (\cosh^2 u)^{\frac{3}{2}} \times \cosh u \, \mathrm{d}u$

$= \int \cosh^3 u \times \cosh u \, \mathrm{d}u$

$= \int \cosh^4 u \, \mathrm{d}u$

Now:

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A = \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore \cosh^4 u$

$= (\cosh^2 u)^2$

$= \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cosh^2 2u + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cosh 4u + \frac{1}{2}\right) + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{8} + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}$

$\therefore I = \int \left(\frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}\right) \mathrm{d}u$

$= \frac{1}{32}\sinh 4u + \frac{1}{4}\sinh 2u + \frac{3}{8}u + c$

$= \frac{1}{32}(2\sinh 2u \cosh 2u) + \frac{1}{4}(2\sinh u \cosh u) + \frac{3}{8}u + c$

$= \frac{1}{16}(2\sinh u \cosh u)(1 + 2\sinh^2 u) + \frac{1}{2}\sinh u \cosh u + \frac{3}{8}u + c$

$= \frac{1}{16}(2\sinh u \cosh u + 4\sinh^3 u \cosh u) + \frac{1}{2}\sinh u \cosh u + \frac{3}{8}u + c$

$= \frac{1}{8}\sinh u \cosh u + \frac{1}{4}\sinh^3 u \cosh u + \frac{1}{2}\sinh u \cosh u + \frac{3}{8}u + c$

$= \frac{5}{8}\sinh u \cosh u + \frac{1}{4}\sinh^3 u \cosh u + \frac{3}{8}u + c$

$= \frac{5}{8}\sinh u \sqrt{\sinh^2 u + 1} + \frac{1}{4}\sinh^3 u \sqrt{\sinh^2 u + 1} + \frac{3}{8}u + c$

$= \frac{5}{8}x\sqrt{x^2 + 1} + \frac{1}{4}x^3\sqrt{x^2 + 1} + \frac{3}{8}\operatorname{arsinh} x + c$

$= \frac{1}{8}x(5 + 2x^2)\sqrt{x^2 + 1} + \frac{3}{8}\operatorname{arsinh} x + c$

So $x = \sinh u$ gives $\int (x^2 + 1)^{\frac{3}{2}} \mathrm{d}x = \frac{1}{8}x(5 + 2x^2)\sqrt{x^2 + 1} + \frac{3}{8}\operatorname{arsinh} x + c$

22) a)

Let $x^2 = \cosh^2 u \Rightarrow x = \cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sinh u \Rightarrow \mathrm{d}x = \sinh u \, \mathrm{d}u$

$\therefore I = \int \sqrt{x^2 - 1} \, \mathrm{d}x$

$= \int \sqrt{\cosh^2 u + 1} \times \sinh u \, \mathrm{d}u$

$= \int \sqrt{\sinh^2 u} \times \sinh u \, \mathrm{d}u$

$= \int \sinh^2 u \, \mathrm{d}u$

$\cosh 2A \equiv 1 + 2\sinh^2 u \Leftrightarrow \sinh^2 u \equiv \frac{1}{2}\cosh 2A - \frac{1}{2}$

$\therefore I = \int \left(\frac{1}{2}\cosh 2u - \frac{1}{2}\right) \mathrm{d}u$

$= \frac{1}{4}\sinh 2u - \frac{1}{2}u + c$

$= \frac{1}{4}(2\sinh u \cosh u) - \frac{1}{2}u + c$

$= \frac{1}{2}\sinh u \cosh u - \frac{1}{2}u + c$

$= \frac{1}{2}\sqrt{\cosh^2 u - 1}\cosh u - \frac{1}{2}u + c$

$= \frac{1}{2}x\sqrt{x^2 - 1} - \frac{1}{2}\operatorname{arcosh} x + c$

So $x = \cosh u$ gives $\int \sqrt{x^2 - 1} \, \mathrm{d}x = \frac{1}{2}x\sqrt{x^2 - 1} - \frac{1}{2}\operatorname{arcosh} x + c$

22) b)

Let $x^2 = \cosh^2 u \Rightarrow x = \cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sinh u \Rightarrow \mathrm{d}x = \sinh u \, \mathrm{d}u$

$\therefore I = \int (x^2 - 1)^{\frac{3}{2}} \mathrm{d}x$

$= \int (\cosh^2 u - 1)^{\frac{3}{2}} \times \sinh u \, \mathrm{d}u$

$= \int (\sinh^2 u)^{\frac{3}{2}} \times \sinh u \, \mathrm{d}u$

$= \int \sinh^3 u \times \sinh u \, \mathrm{d}u$

$= \int \sinh^4 u \, \mathrm{d}u$

Now:

$\cosh 2A \equiv 1 + 2\sinh^2 A \Leftrightarrow \sinh^2 A = \frac{1}{2}\cosh 2A - \frac{1}{2}$

$\therefore \sinh^4 u$

$= (\sinh^2 u)^2$

$= \left(\frac{1}{2}\cosh 2u - \frac{1}{2}\right)^2$

$= \frac{1}{4}\cosh^2 2u - \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cosh 4u + \frac{1}{2}\right) - \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{8} - \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u - \frac{1}{2}\cosh 2u + \frac{3}{8}$

$\therefore I = \int \left(\frac{1}{8}\cosh 4u - \frac{1}{2}\cosh 2u + \frac{3}{8}\right) \mathrm{d}u$

$= \frac{1}{32}\sinh 4u - \frac{1}{4}\sinh 2u + \frac{3}{8}u + c$

$= \frac{1}{32}(2\sinh 2u \cosh 2u) - \frac{1}{4}(2\sinh u \cosh u) + \frac{3}{8}u + c$

$= \frac{1}{16}(2\sinh u \cosh u)(1 + 2\sinh^2 u) - \frac{1}{2}\sinh u \cosh u + \frac{3}{8}u + c$

$= \frac{1}{16}(2\sinh u \cosh u + 4\sinh^3 u \cosh u) - \frac{1}{2}\sinh u \cosh u + \frac{3}{8}u + c$

$= \frac{1}{8}\sinh u \cosh u + \frac{1}{4}\sinh^3 u \cosh u - \frac{1}{2}\sinh u \cosh u + \frac{3}{8}u + c$

$= -\frac{3}{8}\sinh u \cosh u + \frac{1}{4}\sinh^3 u \cosh u + \frac{3}{8}u + c$

$= -\frac{3}{8}\sqrt{\cosh^2 u - 1}\cosh u + \frac{1}{4}\left(\sqrt{\cosh^2 u - 1}\right)^3 \cosh u + \frac{3}{8}u + c$

$= -\frac{3}{8}x\sqrt{x^2 - 1} + \frac{1}{4}x(x^2 - 1)^{\frac{3}{2}} + \frac{3}{8}\operatorname{arcosh} x + c$

$= \frac{1}{8}x\left(-3 + 2(x^2 - 1)\right)\sqrt{x^2 - 1} + \frac{3}{8}\operatorname{arcosh} x + c$

$= \frac{1}{8}x(2x^2 - 5)\sqrt{x^2 - 1} + \frac{3}{8}\operatorname{arcosh} x + c$

So $x = \cosh u$ gives $\int (x^2 - 1)^{\frac{3}{2}}\,\mathrm{d}x = \frac{1}{8}x(2x^2 - 5)\sqrt{x^2 - 1} + \frac{3}{8}\operatorname{arcosh} x + c$

23) a)

Let $x^2 = \tanh^2 u \Rightarrow x = \tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \operatorname{sech}^2 u \, \mathrm{d}u$

$\therefore I = \int \frac{1}{(1-x^2)^2} \mathrm{d}x$

$= \int \frac{1}{(1-\tanh^2 u)^2} \times \operatorname{sech}^2 u \, \mathrm{d}u$

$= \int \frac{\operatorname{sech}^2 u}{(\operatorname{sech}^2 u)^2} \mathrm{d}u$

$= \int \frac{1}{\operatorname{sech}^2 u} \mathrm{d}u$

$= \int \cosh^2 u \, \mathrm{d}u$

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A \equiv \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore I = \int \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right) \mathrm{d}u$

$= \frac{1}{4}\sinh 2u + \frac{1}{2}u + c$

$= \frac{1}{4}(2\sinh u \cosh u) + \frac{1}{2}u + c$

$= \frac{1}{2}\sinh u \cosh u + \frac{1}{2}u + c$

If $\tanh u = x$

$\Rightarrow \frac{\sinh u}{\cosh u} = x$

$\Rightarrow \sinh u = x \cosh u$

Use the identity $\cosh^2 A - \sinh^2 A \equiv 1$

$\therefore \cosh^2 u - (x\cosh u)^2 = 1$

$\Rightarrow \cosh^2 u - x^2 \cosh^2 u = 1$

$\Rightarrow \cosh^2 u \, (1 - x^2) = 1$

$\Rightarrow \cosh^2 u = \frac{1}{1-x^2}$

$\Rightarrow \cosh u = \frac{1}{\sqrt{1-x^2}}$ as $\cosh u > 0$

$\Rightarrow \sinh u = \frac{x}{\sqrt{1-x^2}}$

$\therefore I = \frac{1}{2} \times \frac{x}{\sqrt{1-x^2}} \times \frac{1}{\sqrt{1-x^2}} + \frac{1}{2}\operatorname{artanh} x + c$

$= \frac{x}{2(1-x^2)} + \frac{1}{2}\operatorname{artanh} x + c$

So $x = \tanh u$ gives $\int \frac{1}{(1-x^2)^2} \mathrm{d}x = \frac{x}{2(1-x^2)} + \frac{1}{2}\operatorname{artanh} x + c$

23) b)

Let $9x^2 = \tanh^2 u \Rightarrow 3x = \tanh u \Rightarrow x = \frac{1}{3}\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{3}\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \frac{1}{3}\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore I = \int \frac{1}{(1-9x^2)^2}\,\mathrm{d}x$

$= \int \frac{1}{(1-\tanh^2 u)^2} \times \frac{1}{3}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{1}{3}\int \frac{\operatorname{sech}^2 u}{(\operatorname{sech}^2 u)^2}\,\mathrm{d}u$

$= \frac{1}{3}\int \frac{1}{\operatorname{sech}^2 u}\,\mathrm{d}u$

$= \frac{1}{3}\int \cosh^2 u\,\mathrm{d}u$

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A \equiv \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore I = \frac{1}{3}\int \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)\mathrm{d}u$

$= \int \left(\frac{1}{6}\cosh 2u + \frac{1}{6}\right)\mathrm{d}u$

$= \frac{1}{12}\sinh 2u + \frac{1}{6}u + c$

$= \frac{1}{12}(2\sinh u \cosh u) + \frac{1}{6}u + c$

$= \frac{1}{6}\sinh u \cosh u + \frac{1}{6}u + c$

If $\tanh u = 3x$

$\Rightarrow \frac{\sinh u}{\cosh u} = 3x$

$\Rightarrow \sinh u = 3x\cosh u$

Use the identity $\cosh^2 A - \sinh^2 A \equiv 1$

$\therefore \cosh^2 u - (3x\cosh u)^2 = 1$

$\Rightarrow \cosh^2 u - 9x^2\cosh^2 u = 1$

$\Rightarrow \cosh^2 u\,(1 - 9x^2) = 1$

$\Rightarrow \cosh^2 u = \frac{1}{1-9x^2}$

$\Rightarrow \cosh u = \frac{1}{\sqrt{1-9x^2}}$ as $\cosh u > 0$

$\Rightarrow \sinh u = \frac{3x}{\sqrt{1-9x^2}}$

$\therefore I = \frac{1}{6} \times \frac{3x}{\sqrt{1-9x^2}} \times \frac{1}{\sqrt{1-9x^2}} + \frac{1}{6}\operatorname{artanh} 3x + c$

$= \frac{x}{2(1-9x^2)} + \frac{1}{6}\operatorname{artanh} 3x + c$

So $x = \frac{1}{3}\tanh u$ gives $\int \frac{1}{(1-9x^2)^2}\,\mathrm{d}x = \frac{x}{2(1-9x^2)} + \frac{1}{6}\operatorname{artanh} 3x + c$

23) c)

Let $x^2 = \tanh^2 u \Rightarrow x = \tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \operatorname{sech}^2 u \Rightarrow \mathrm{d}x = \operatorname{sech}^2 u \, \mathrm{d}u$

$\therefore I = \int \frac{1}{(1-x^2)^3} \mathrm{d}x$

$= \int \frac{1}{(1-\tanh^2 u)^3} \times \operatorname{sech}^2 u \, \mathrm{d}u$

$= \int \frac{\operatorname{sech}^2 u}{(\operatorname{sech}^2 u)^3} \mathrm{d}u$

$= \int \frac{1}{(\operatorname{sech}^2 u)^2} \mathrm{d}u$

$= \int \frac{1}{\operatorname{sech}^4 u} \mathrm{d}u$

$= \int \cosh^4 u \, \mathrm{d}u$

Now:

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A = \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore \cosh^4 u$

$= (\cosh^2 u)^2$

$= \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cosh^2 2u + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cosh 4u + \frac{1}{2}\right) + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{8} + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}$

$\therefore I = \int \left(\frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}\right) \mathrm{d}u$

$= \frac{1}{32}\sinh 4u + \frac{1}{4}\sinh 2u + \frac{3}{8}u + c$

$= \frac{1}{32}(2\sinh 2u \cosh 2u) + \frac{1}{4}(2\sinh u \cosh u) + \frac{3}{8}u + c$

$= \frac{1}{16}\sinh 2u \cosh 2u + \frac{1}{2}\sinh u \cosh u + \frac{3}{8}u + c$

$= \frac{1}{16}(2\sinh u \cosh u)(2\cosh^2 u - 1) + \frac{1}{2}\sinh u \cosh u + \frac{3}{8}u + c$

$= \frac{1}{4}\sinh u \cosh^3 u - \frac{1}{8}\sinh u \cosh u + \frac{1}{2}\sinh u \cosh u + \frac{3}{8}u + c$

$= \frac{1}{4}\sinh u \cosh^3 u + \frac{3}{8}\sinh u \cosh u + \frac{3}{8}u + c$

If $\tanh u = x$

$\Rightarrow \frac{\sinh u}{\cosh u} = x$

$\Rightarrow \sinh u = x \cosh u$

Use the identity $\cosh^2 A - \sinh^2 A \equiv 1$

$\therefore \cosh^2 u - (x \cosh u)^2 = 1$

$\Rightarrow \cosh^2 u - x^2 \cosh^2 u = 1$

$\Rightarrow \cosh^2 u\,(1 - x^2) = 1$

$\Rightarrow \cosh^2 u = \frac{1}{1-x^2}$

$\Rightarrow \cosh u = \frac{1}{\sqrt{1-x^2}}$ as $\cosh u > 0$

$\Rightarrow \sinh u = \frac{x}{\sqrt{1-x^2}}$

$\therefore I = \frac{1}{4} \times \frac{x}{\sqrt{1-x^2}} \times \left(\frac{1}{\sqrt{1-x^2}}\right)^3 + \frac{3}{8} \times \frac{x}{\sqrt{1-x^2}} \times \frac{1}{\sqrt{1-x^2}} + \frac{3}{8}\operatorname{artanh} x + c$

$= \frac{x}{4(1-x^2)^2} + \frac{3x}{8(1-x^2)} + \frac{3}{8}\operatorname{artanh} x + c$

$= \frac{2x}{8(1-x^2)^2} + \frac{3x(1-x^2)}{8(1-x^2)^2} + \frac{3}{8}\operatorname{artanh} x + c$

$= \frac{2x+3x(1-x^2)}{8(1-x^2)^2} + \frac{3}{8}\operatorname{artanh} x + c$

$= \frac{5x-3x^3}{8(1-x^2)^2} + \frac{3}{8}\operatorname{artanh} x + c$

So $x = \tanh u$ gives $\int \frac{1}{(1-x^2)^3}\,\mathrm{d}x = \frac{5x-3x^3}{8(1-x^2)^2} + \frac{3}{8}\operatorname{artanh} x + c$

23) d)

Let $x^2 = 4\tanh^2 u \Rightarrow x = 2\tanh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\operatorname{sech}^2 u \Rightarrow \mathrm{d}x = 2\operatorname{sech}^2 u\,\mathrm{d}u$

$\therefore I = \int \frac{1}{(4-x^2)^3}\,\mathrm{d}x$

$= \int \frac{1}{(4-4\tanh^2 u)^3} \times 2\operatorname{sech}^2 u\,\mathrm{d}u$

$= \int \frac{1}{\left(4(1-\tanh^2 u)\right)^3} \times 2\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{1}{32}\int \frac{\operatorname{sech}^2 u}{(\operatorname{sech}^2 u)^3}\,\mathrm{d}u$

$= \frac{1}{32}\int \frac{1}{(\operatorname{sech}^2 u)^2}\,\mathrm{d}u$

$= \frac{1}{32}\int \frac{1}{\operatorname{sech}^4 u}\,\mathrm{d}u$

$= \frac{1}{32}\int \cosh^4 u\,\mathrm{d}u$

Now:

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A = \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore \cosh^4 u$

$= (\cosh^2 u)^2$

$= \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cosh^2 2u + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cosh 4u + \frac{1}{2}\right) + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{8} + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}$

$\therefore I = \frac{1}{32}\int \left(\frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}\right)\mathrm{d}u$

$= \int \left(\frac{1}{256}\cosh 4u + \frac{1}{64}\cosh 2u + \frac{3}{256}\right)\mathrm{d}u$

$= \frac{1}{1024}\sinh 4u + \frac{1}{128}\sinh 2u + \frac{3}{256}u + c$

$= \frac{1}{1024}(2\sinh 2u\cosh 2u) + \frac{1}{128}(2\sinh u\cosh u) + \frac{3}{256}u + c$

$= \frac{1}{512}\sinh 2u\cosh 2u + \frac{1}{64}\sinh u\cosh u + \frac{3}{256}u + c$

$= \frac{1}{512}(2\sinh u\cosh u)(2\cosh^2 u - 1) + \frac{1}{64}\sinh u\cosh u + \frac{3}{256}u + c$

$= \frac{1}{128}\sinh u\cosh^3 u - \frac{1}{256}\sinh u\cosh u + \frac{1}{64}\sinh u\cosh u + \frac{3}{256}u + c$

$= \frac{1}{128}\sinh u\cosh^3 u + \frac{3}{256}\sinh u\cosh u + \frac{3}{256}u + c$

If $\tanh u = \frac{1}{2}x$

$\Rightarrow \frac{\sinh u}{\cosh u} = \frac{1}{2}x$

$\Rightarrow \sinh u = \frac{1}{2}x\cosh u$

Use the identity $\cosh^2 A - \sinh^2 A \equiv 1$

$\therefore \cosh^2 u - \left(\frac{1}{2}x\cosh u\right)^2 = 1$

$\Rightarrow \cosh^2 u - \frac{1}{4}x^2\cosh^2 u = 1$

$\Rightarrow \cosh^2 u\left(1 - \frac{1}{4}x^2\right) = 1$

$\Rightarrow \cosh^2 u = \frac{1}{1-\frac{1}{4}x^2} = \frac{4}{4-x^2}$

$\Rightarrow \cosh u = \frac{2}{\sqrt{4-x^2}}$ as $\cosh u > 0$

$\Rightarrow \sinh u = \frac{x}{\sqrt{4-x^2}}$

$\therefore I = \frac{1}{128} \times \frac{x}{\sqrt{4-x^2}} \times \left(\frac{2}{\sqrt{4-x^2}}\right)^3 + \frac{3}{256} \times \frac{x}{\sqrt{4-x^2}} \times \frac{2}{\sqrt{4-x^2}} + \frac{3}{256}\operatorname{artanh}\left(\frac{1}{2}x\right) + c$

$= \frac{x}{16(4-x^2)^2} + \frac{3x}{128(4-x^2)} + \frac{3}{256}\operatorname{artanh}\left(\frac{1}{2}x\right) + c$

$= \frac{8x}{128(4-x^2)^2} + \frac{3x(4-x^2)}{128(4-x^2)^2} + \frac{3}{256}\operatorname{artanh}\left(\frac{1}{2}x\right) + c$

$= \frac{8x+3x(4-x^2)}{128(1-x^2)^2} + \frac{3}{256}\operatorname{artanh}\left(\frac{1}{2}x\right) + c$

$= \frac{20x-3x^3}{128(1-x^2)^2} + \frac{3}{256}\operatorname{artanh}\left(\frac{1}{2}x\right) + c$

So $x = 2\tanh u$ gives $\int \frac{1}{(4-x^2)^3}\,\mathrm{d}x = \frac{20x-3x^3}{128(1-x^2)^2} + \frac{3}{256}\operatorname{artanh}\left(\frac{1}{2}x\right) + c$

24) a)

Let $25x^2 = \sinh^2 u \Rightarrow 5x = \sinh u \Rightarrow x = \frac{1}{5}\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{5}\cosh u \Rightarrow \mathrm{d}x = \frac{1}{5}\cosh u \,\mathrm{d}u$

$\therefore I = \int \sqrt{25x^2 + 1}\,\mathrm{d}x$

$= \int \sqrt{\sinh^2 u + 1} \times \frac{1}{5}\cosh u \,\mathrm{d}u$

$= \frac{1}{5}\int \sqrt{\cosh^2 u} \times \cosh u \,\mathrm{d}u$

$= \frac{1}{5}\int \cosh^2 u \,\mathrm{d}u$

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A \equiv \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore I = \frac{1}{5}\int \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)\mathrm{d}u$

$= \int \left(\frac{1}{10}\cosh 2u + \frac{1}{10}\right)\mathrm{d}u$

$= \frac{1}{20}\sinh 2u + \frac{1}{10}u + c$

$= \frac{1}{20}(2\sinh u \cosh u) + \frac{1}{10}u + c$

$= \frac{1}{10}\sinh u \cosh u + \frac{1}{10}u + c$

$= \frac{1}{10}\sinh u \sqrt{\sinh^2 u + 1} + \frac{1}{10}u + c$

$= \frac{1}{10}(5x)\sqrt{(5x)^2 + 1} + \frac{1}{10}\operatorname{arsinh} 5x + c$

$= \frac{1}{2}x\sqrt{25x^2 + 1} + \frac{1}{10}\operatorname{arsinh} 5x + c$

So $x = \frac{1}{5}\sinh u$ gives $\int \sqrt{25x^2 + 1}\,\mathrm{d}x = \frac{1}{2}x\sqrt{25x^2 + 1} + \frac{1}{10}\operatorname{arsinh} 5x + c$

24) b)

Let $x^2 = 9\sinh^2 u \Rightarrow x = 3\sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 3\cosh u \Rightarrow \mathrm{d}x = 3\cosh u\,\mathrm{d}u$

$\therefore I = \int (x^2+9)^{\frac{3}{2}}\,\mathrm{d}x$

$= \int (9\sinh^2 u + 9)^{\frac{3}{2}} \times 3\cosh u\,\mathrm{d}u$

$= \int \left(9(\sinh^2 u + 1)\right)^{\frac{3}{2}} \times 3\cosh u\,\mathrm{d}u$

$= 81\int (\cosh^2 u)^{\frac{3}{2}} \times \cosh u\,\mathrm{d}u$

$= 81\int \cosh^3 u \times \cosh u\,\mathrm{d}u$

$= 81\int \cosh^4 u\,\mathrm{d}u$

Now:

$\cosh 2A \equiv 2\cosh^2 A - 1 \Leftrightarrow \cosh^2 A = \frac{1}{2}\cosh 2A + \frac{1}{2}$

$\therefore \cosh^4 u$

$= (\cosh^2 u)^2$

$= \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)^2$

$= \frac{1}{4}\cosh^2 2u + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cosh 4u + \frac{1}{2}\right) + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{8} + \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}$

$\therefore I = 81\int \left(\frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}\right)\mathrm{d}u$

$= \int \left(\frac{81}{8}\cosh 4u + \frac{81}{2}\cosh 2u + \frac{243}{8}\right)\mathrm{d}u$

$= \frac{81}{32}\sinh 4u + \frac{81}{4}\sinh 2u + \frac{243}{8}u + c$

$= \frac{81}{32}(2\sinh 2u\cosh 2u) + \frac{81}{4}(2\sinh u\cosh u) + \frac{243}{8}u + c$

$= \frac{81}{16}(2\sinh u\cosh u)(1 + 2\sinh^2 u) + \frac{81}{2}\sinh u\cosh u + \frac{243}{8}u + c$

$= \frac{81}{16}(2\sinh u\cosh u + 4\sinh^3 u\cosh u) + \frac{81}{2}\sinh u\cosh u + \frac{243}{8}u + c$

$= \frac{81}{8}\sinh u\cosh u + \frac{81}{4}\sinh^3 u\cosh u + \frac{81}{2}\sinh u\cosh u + \frac{243}{8}u + c$

$= \frac{405}{8}\sinh u\cosh u + \frac{81}{4}\sinh^3 u\cosh u + \frac{243}{8}u + c$

$= \frac{405}{8}\sinh u\sqrt{\sinh^2 u + 1} + \frac{81}{4}\sinh^3 u\sqrt{\sinh^2 u + 1} + \frac{243}{8}u + c$

$= \frac{405}{8}\left(\frac{1}{3}x\right)\sqrt{\left(\frac{1}{3}x\right)^2 + 1} + \frac{81}{4}\left(\frac{1}{3}x\right)^3\sqrt{\left(\frac{1}{3}x\right)^2 + 1} + \frac{243}{8}\operatorname{arsinh}\left(\frac{1}{3}x\right) + c$

$= \frac{135}{8}x\sqrt{\frac{1}{9}x^2+1} + \frac{3}{4}x^3\sqrt{\frac{1}{9}x^2+1} + \frac{243}{8}\operatorname{arsinh}\left(\frac{1}{3}x\right) + c$

$= \frac{135}{8}x\sqrt{\frac{1}{9}(x^2+9)} + \frac{3}{4}x^3\sqrt{\frac{1}{9}(x^2+9)} + \frac{243}{8}\operatorname{arsinh}\left(\frac{1}{3}x\right) + c$

$= \frac{45}{8}x\sqrt{x^2+9} + \frac{1}{4}x^3\sqrt{x^2+9} + \frac{243}{8}\operatorname{arsinh}\left(\frac{1}{3}x\right) + c$

$= \frac{1}{8}x(45+2x^2)\sqrt{x^2+9} + \frac{243}{8}\operatorname{arsinh}\left(\frac{1}{3}x\right) + c$

So $x = 3\sinh u$ gives $\int (x^2+9)^{\frac{3}{2}}\,\mathrm{d}x = \frac{1}{8}x(45+2x^2)\sqrt{x^2+9} + \frac{243}{8}\operatorname{arsinh}\left(\frac{1}{3}x\right) + c$

24) c)

Let $x^2 = 36\cosh^2 u \Rightarrow x = 6\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 6\sinh u \Rightarrow \mathrm{d}x = 6\sinh u\,\mathrm{d}u$

$\therefore I = \int \sqrt{x^2-36}\,\mathrm{d}x$

$= \int \sqrt{36\cosh^2 u - 36} \times 6\sinh u\,\mathrm{d}u$

$= \int \sqrt{36(\cosh^2 u - 1)} \times 6\sinh u\,\mathrm{d}u$

$= 36\int \sqrt{\sinh^2 u} \times \sinh u\,\mathrm{d}u$

$= 36\int \sinh^2 u\,\mathrm{d}u$

$\cosh 2A \equiv 1 + 2\sinh^2 u \Leftrightarrow \sinh^2 u \equiv \frac{1}{2}\cosh 2A - \frac{1}{2}$

$\therefore I = 36\int \left(\frac{1}{2}\cosh 2u - \frac{1}{2}\right)\mathrm{d}u$

$= \int (18\cosh 2u - 18)\,\mathrm{d}u$

$= 9\sinh 2u - 18u + c$

$= 9(2\sinh u\cosh u) - 18u + c$

$= 18\sinh u\cosh u - 18u + c$

$= 18\sqrt{\cosh^2 u - 1}\cosh u - 18u + c$

$= 18\left(\frac{1}{6}x\right)\sqrt{\left(\frac{1}{6}x\right)^2 - 1} - 18\operatorname{arcosh}\left(\frac{1}{6}x\right) + c$

$= 3x\sqrt{\frac{1}{36}x^2 - 1} - 18\operatorname{arcosh}\left(\frac{1}{6}x\right) + c$

$= 3x\sqrt{\frac{1}{36}(x^2-36)} - 18\operatorname{arcosh}\left(\frac{1}{6}x\right) + c$

$= \frac{1}{2}x\sqrt{x^2-36} - 18\operatorname{arcosh}\left(\frac{1}{6}x\right) + c$

So $x = 6\cosh u$ gives $\int \sqrt{x^2-36}\,\mathrm{d}x = \frac{1}{2}x\sqrt{x^2-36} - 18\operatorname{arcosh}\left(\frac{1}{6}x\right) + c$

24) d)

Let $81x^2 = \cosh^2 u \Rightarrow 9x = \cosh u \Rightarrow x = \frac{1}{9}\cosh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{9}\sinh u \Rightarrow \mathrm{d}x = \frac{1}{9}\sinh u\,\mathrm{d}u$

$\therefore I = \int (81x^2 - 1)^{\frac{3}{2}}\,\mathrm{d}x$

$= \int (\cosh^2 u - 1)^{\frac{3}{2}} \times \frac{1}{9}\sinh u\,\mathrm{d}u$

$= \frac{1}{9}\int (\sinh^2 u)^{\frac{3}{2}} \times \sinh u\,\mathrm{d}u$

$= \frac{1}{9}\int \sinh^3 u \times \sinh u\,\mathrm{d}u$

$= \frac{1}{9}\int \sinh^4 u\,\mathrm{d}u$

Now:

$\cosh 2A \equiv 1 + 2\sinh^2 A \Leftrightarrow \sinh^2 A = \frac{1}{2}\cosh 2A - \frac{1}{2}$

$\therefore \sinh^4 u$

$= (\sinh^2 u)^2$

$= \left(\frac{1}{2}\cosh 2u - \frac{1}{2}\right)^2$

$= \frac{1}{4}\cosh^2 2u - \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{4}\left(\frac{1}{2}\cosh 4u + \frac{1}{2}\right) - \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u + \frac{1}{8} - \frac{1}{2}\cosh 2u + \frac{1}{4}$

$= \frac{1}{8}\cosh 4u - \frac{1}{2}\cosh 2u + \frac{3}{8}$

$\therefore I = \frac{1}{9}\int \left(\frac{1}{8}\cosh 4u - \frac{1}{2}\cosh 2u + \frac{3}{8}\right)\mathrm{d}u$

$= \int \left(\frac{1}{72}\cosh 4u - \frac{1}{18}\cosh 2u + \frac{1}{24}\right)\mathrm{d}u$

$= \frac{1}{288}\sinh 4u - \frac{1}{36}\sinh 2u + \frac{1}{24}u + c$

$= \frac{1}{288}(2\sinh 2u \cosh 2u) - \frac{1}{36}(2\sinh u \cosh u) + \frac{1}{24}u + c$

$= \frac{1}{144}(2\sinh u \cosh u)(1 + 2\sinh^2 u) - \frac{1}{18}\sinh u \cosh u + \frac{1}{24}u + c$

$= \frac{1}{144}(2\sinh u \cosh u + 4\sinh^3 u \cosh u) - \frac{1}{18}\sinh u \cosh u + \frac{1}{24}u + c$

$= \frac{1}{72}\sinh u \cosh u + \frac{1}{36}\sinh^3 u \cosh u - \frac{1}{18}\sinh u \cosh u + \frac{1}{24}u + c$

$= -\frac{1}{24}\sinh u \cosh u + \frac{1}{36}\sinh^3 u \cosh u + \frac{1}{24}u + c$

$= -\frac{1}{24}\sqrt{\cosh^2 u - 1}\cosh u + \frac{1}{36}\left(\sqrt{\cosh^2 u - 1}\right)^3 \cosh u + \frac{1}{24}u + c$

$= -\frac{1}{24}(9x)\sqrt{(9x)^2 - 1} + \frac{1}{36}(9x)((9x)^2 - 1)^{\frac{3}{2}} + \frac{1}{24}\operatorname{arcosh} 9x + c$

$= -\frac{3}{8}x\sqrt{81x^2-1} + \frac{1}{4}x(81x^2-1)^{\frac{3}{2}} + \frac{1}{24}\operatorname{arcosh} 9x + c$

$= \frac{1}{8}x\left(-3 + 2(81x^2-1)\right)\sqrt{81x^2-1} + \frac{1}{24}\operatorname{arcosh} 9x + c$

$= \frac{1}{8}x(162x^2-5)\sqrt{81x^2-1} + \frac{1}{24}\operatorname{arcosh} 9x + c$

So $x = \frac{1}{9}\cosh u$ gives $\int (81x^2-1)^{\frac{3}{2}}\,\mathrm{d}x = \frac{1}{8}x(162x^2-5)\sqrt{81x^2-1} + \frac{1}{24}\operatorname{arcosh} 9x + c$

25) a)

$x^2 + 4x + 3$

$\equiv (x+2)^2 - 4 + 3$

$\equiv (x+2)^2 - 1$

25) b)

$x^2 - \frac{10}{3}x + \frac{34}{9}$

$\equiv \left(x - \frac{5}{3}\right)^2 - \frac{25}{9} + \frac{34}{9}$

$\equiv \left(x - \frac{5}{3}\right)^2 + 1$

25) c)

$4x^2 + 24x + 37$

$\equiv 4(x^2 + 6x) + 37$

$\equiv 4((x+3)^2 - 9) + 37$

$\equiv 4(x+3)^2 - 36 + 37$

$\equiv 4(x+3)^2 + 1$

25) d)

$9x^2 - 18x + 8$

$\equiv 9(x^2 - 2x) + 8$

$\equiv 9((x-1)^2 - 1) + 8$

$\equiv 9(x-1)^2 - 9 + 8$

$\equiv 9(x-1)^2 - 1$

25) e)

$x^2 + 6x + 18$
$\equiv (x+3)^2 - 9 + 18$
$\equiv (x+3)^2 + 9$

25) f)

$9x^2 - 36x + 32$
$\equiv 9(x^2 - 4x) + 32$
$\equiv 9((x-2)^2 - 4) + 32$
$\equiv 9(x-2)^2 - 36 + 32$
$\equiv 9(x-2)^2 - 4$

26) a)

$x^2 + 20x + 101$
$\equiv (x+10)^2 - 100 + 101$
$\equiv (x+10)^2 + 1$

26) b)

Let $x + 10 = \sinh u \Rightarrow x = \sinh u - 10 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u \ \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{x^2+20x+101}} \mathrm{d}x = \int \frac{1}{\sqrt{(x+10)^2+1}} \mathrm{d}x = \int \frac{1}{\sqrt{\sinh^2 u+1}} \times \cosh u \, \mathrm{d}u$

$= \int \frac{\cosh u}{\sqrt{\cosh^2 u}} \mathrm{d}u$
$= \int \frac{\cosh u}{\cosh u} \mathrm{d}u$
$= \int 1 \, \mathrm{d}u$
$= u + c$
$= \operatorname{arsinh}(x+10) + c$

27) a)

$x^2 - 14x + 48$
$\equiv (x-7)^2 - 49 + 48$
$\equiv (x-7)^2 - 1$

27) b)

Let $x-7=\cosh u \Rightarrow x=\cosh u+7 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u}=\sinh u \Rightarrow \mathrm{d}x=\sinh u\ \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{x^2-14x+48}}\mathrm{d}x=\int \frac{1}{\sqrt{(x-7)^2-1}}\mathrm{d}x=\int \frac{1}{\sqrt{\cosh^2 u-1}}\times \sinh u\,\mathrm{d}u$

$=\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$=\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$=\int 1\,\mathrm{d}u$

$=u+c$

$=\operatorname{arcosh}(x-7)+c$

28) a)

$\int \frac{1}{\sqrt{x^2-10x+26}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{(x-5)^2-25+26}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{(x-5)^2+1}}\mathrm{d}x$

$=\operatorname{arsinh}(x-5)+c$

28) b)

$\int \frac{1}{\sqrt{x^2+\frac{2}{3}x-\frac{8}{9}}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{\left(x+\frac{1}{3}\right)^2-\frac{1}{9}-\frac{8}{9}}}\mathrm{d}x$

$=\int \frac{1}{\sqrt{\left(x+\frac{1}{3}\right)^2-1}}\mathrm{d}x$

$=\operatorname{arcosh}\left(x+\frac{1}{3}\right)+c$

29)

Mean Value:

$$= \frac{1}{\frac{2}{5}-\left(-\frac{2}{5}\right)} \int_{-\frac{2}{5}}^{\frac{2}{5}} \frac{1}{\sqrt{x^2-\frac{4}{5}x+\frac{29}{25}}}\,\mathrm{d}x$$

$$= \frac{5}{4} \int_{-\frac{2}{5}}^{\frac{2}{5}} \frac{1}{\sqrt{\left(x-\frac{2}{5}\right)^2-\frac{4}{25}+\frac{29}{25}}}\,\mathrm{d}x$$

$$= \frac{5}{4} \int_{-\frac{2}{5}}^{\frac{2}{5}} \frac{1}{\sqrt{\left(x-\frac{2}{5}\right)^2+1}}\,\mathrm{d}x$$

$$= \frac{5}{4} \left[\operatorname{arsinh}\left(x-\frac{2}{5}\right)\right]_{-\frac{2}{5}}^{\frac{2}{5}}$$

$$= \frac{5}{4} \left(\operatorname{arsinh} 0 - \operatorname{arsinh}\left(-\frac{4}{5}\right)\right)$$

$$= \frac{5}{4} \left(0 - \ln\left(-\frac{4}{5}+\sqrt{\left(-\frac{4}{5}\right)^2+1}\right)\right)$$

$$= -\frac{5}{4} \ln\left(-\frac{4}{5}+\frac{\sqrt{41}}{5}\right)$$

$$= -\frac{5}{4} \ln\left(\frac{-4+\sqrt{41}}{5}\right)$$

30) a)

$$4x^2 - 40x + 101$$
$$\equiv 4(x^2 - 10x) + 101$$
$$\equiv 4((x-5)^2 - 25) + 101$$
$$\equiv 4(x-5)^2 - 100 + 101$$
$$\equiv 4(x-5)^2 + 1$$

30) b)

Let $2x - 10 = \sinh u$

$\Rightarrow x = \frac{1}{2}\sinh u + 5$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{2}\cosh u$

$\Rightarrow \mathrm{d}x = \frac{1}{2}\cosh u \ \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{4x^2-40x+101}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{4(x-5)^2+1}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{(2x-10)^2+1}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{\sinh^2 u+1}} \times \frac{1}{2}\cosh u \,\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \frac{1}{2}\int 1 \,\mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\mathrm{arsinh}(2x - 10) + c$

31) a)

$16x^2 + 96x + 143$

$\equiv 16(x^2 + 6x) + 143$

$\equiv 16((x + 3)^2 - 9) + 143$

$\equiv 16(x + 3)^2 - 144 + 143$

$\equiv 16(x + 3)^2 - 1$

31) b)

Let $4x + 12 = \cosh u$

$\Rightarrow x = \frac{1}{4}\cosh u - 3$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{4}\sinh u$

$\Rightarrow \mathrm{d}x = \frac{1}{4}\sinh u \ \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{16x^2+96x+143}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{16(x+3)^2-1}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{(4x+12)^2-1}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{\cosh^2 u-1}} \times \frac{1}{4}\sinh u \, \mathrm{d}u$

$= \frac{1}{4}\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \frac{1}{4}\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \frac{1}{4}\int 1 \, \mathrm{d}u$

$= \frac{1}{4}u + c$

$= \frac{1}{4}\operatorname{arcosh}(4x + 12) + c$

32) a)

$\int \frac{1}{\sqrt{9x^2-144x+575}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9(x^2-16x)+575}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9((x-8)^2-64)+575}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9(x-8)^2-576+575}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{(3x-24)^2-1}}\mathrm{d}x$

$= \frac{1}{3}\operatorname{arcosh}(3x - 24) + c$

32) b)

$$\int \frac{1}{\sqrt{2x^2+12x+19}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{2(x^2+6x)+19}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{2((x+3)^2-9)+19}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{2(x+3)^2-18+19}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{\left(\sqrt{2}x+3\sqrt{2}\right)^2+1}}\,\mathrm{d}x$$
$$= \frac{1}{\sqrt{2}}\operatorname{arsinh}\left(\sqrt{2}x+3\sqrt{2}\right)+c$$
$$= \frac{\sqrt{2}}{2}\operatorname{arsinh}\left(\sqrt{2}x+3\sqrt{2}\right)+c$$

33) a)

$$4x^2-56x+205$$
$$\equiv 4(x^2-14x)+205$$
$$\equiv 4((x-7)^2-49)+205$$
$$\equiv 4(x-7)^2-196+205$$
$$\equiv 4(x-7)^2+9$$

33) b)

Let $\frac{2}{3}(x-7) = \sinh u$

$\Rightarrow x = \frac{3}{2}\sinh u + 7$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{2}\cosh u$

$\Rightarrow \mathrm{d}x = \frac{3}{2}\cosh u \,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{4x^2-56x+205}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{4(x-7)^2+9}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{9\left(\frac{4}{9}(x-7)^2+1\right)}}\mathrm{d}x$

$= \frac{1}{3}\int \frac{1}{\sqrt{\frac{4}{9}(x-7)^2+1}}\mathrm{d}x$

$= \frac{1}{3}\int \frac{1}{\sqrt{\left(\frac{2}{3}x-\frac{14}{3}\right)^2+1}}\mathrm{d}x$

$= \frac{1}{3}\int \frac{1}{\sqrt{\sinh^2 u+1}} \times \frac{3}{2}\cosh u \,\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\mathrm{d}u$

$= \frac{1}{2}\int \frac{\cosh u}{\cosh u}\mathrm{d}u$

$= \frac{1}{2}\int 1 \,\mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\mathrm{arsinh}\left(\frac{2}{3}(x-7)\right) + c$

$= \frac{1}{2}\mathrm{arsinh}\left(\frac{2}{3}x - \frac{14}{3}\right) + c$

34) a)

$3x^2 + 6x - 1$

$\equiv 3(x^2 + 2x) - 1$

$\equiv 3((x+1)^2 - 1) - 1$

$\equiv 3(x+1)^2 - 3 - 1$

$\equiv 3(x+1)^2 - 4$

34) b)

Let $\frac{\sqrt{3}}{2}(x+1) = \cosh u$

$\Rightarrow x = \frac{2}{\sqrt{3}}\cosh u - 1$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{2\sqrt{3}}{3}\sinh u$

$\Rightarrow \mathrm{d}x = \frac{2\sqrt{3}}{3}\sinh u \,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{3x^2+6x-1}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{3(x+1)^2-4}}\mathrm{d}x$

$= \int \frac{1}{\sqrt{4\left(\frac{3}{4}(x+1)^2-1\right)}}\mathrm{d}x$

$= \frac{1}{2}\int \frac{1}{\sqrt{\frac{3}{4}(x+1)^2-1}}\mathrm{d}x$

$= \frac{1}{2}\int \frac{1}{\sqrt{\left(\frac{\sqrt{3}}{2}x+\frac{\sqrt{3}}{2}\right)^2-1}}\mathrm{d}x$

$= \frac{1}{2}\int \frac{1}{\sqrt{\cosh^2 u-1}} \times \frac{2\sqrt{3}}{3}\sinh u \,\mathrm{d}u$

$= \frac{\sqrt{3}}{3}\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\mathrm{d}u$

$= \frac{\sqrt{3}}{3}\int \frac{\sinh u}{\sinh u}\mathrm{d}u$

$= \frac{\sqrt{3}}{3}\int 1 \,\mathrm{d}u$

$= \frac{\sqrt{3}}{3}u + c$

$= \frac{\sqrt{3}}{3}\operatorname{arcosh}\left(\frac{\sqrt{3}}{2}(x+1)\right) + c$

$= \frac{\sqrt{3}}{3}\operatorname{arcosh}\left(\frac{\sqrt{3}}{2}x+\frac{\sqrt{3}}{2}\right) + c$

35) a)

$$\int \frac{1}{\sqrt{25x^2-100x+91}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{25(x^2-4x)+91}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{25((x-2)^2-4)+91}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{25(x-2)^2-100+91}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{25(x-2)^2-9}}\,\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{9\left(\frac{25}{9}(x-2)^2-1\right)}}\,\mathrm{d}x$$
$$= \frac{1}{3}\int \frac{1}{\sqrt{\frac{25}{9}(x-2)^2-1}}\,\mathrm{d}x$$
$$= \frac{1}{3}\int \frac{1}{\sqrt{\left(\frac{5}{3}x-\frac{10}{3}\right)^2-1}}\,\mathrm{d}x$$
$$= \frac{1}{3}\times\frac{1}{\frac{5}{3}}\times \operatorname{arcosh}\left(\frac{5}{3}x-\frac{10}{3}\right)+c$$
$$= \frac{1}{5}\operatorname{arcosh}\left(\frac{5}{3}x-\frac{10}{3}\right)+c$$

35) b)

$$\int \frac{1}{\sqrt{5x^2+10\sqrt{5}x+29}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{5\left(x^2+2\sqrt{5}x\right)+29}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{5\left(\left(x+\sqrt{5}\right)^2-5\right)+29}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{5\left(x+\sqrt{5}\right)^2-25+29}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{5\left(x+\sqrt{5}\right)^2+4}}\,\mathrm{d}x$$

$$= \int \frac{1}{\sqrt{4\left(\frac{5}{4}\left(x+\sqrt{5}\right)^2+1\right)}}\,\mathrm{d}x$$

$$= \frac{1}{2}\int \frac{1}{\sqrt{\frac{5}{4}\left(x+\sqrt{5}\right)^2+1}}\,\mathrm{d}x$$

$$= \frac{1}{2}\int \frac{1}{\sqrt{\left(\frac{\sqrt{5}}{2}x+\frac{5}{2}\right)^2+1}}\,\mathrm{d}x$$

$$= \frac{1}{2}\times\frac{1}{\frac{\sqrt{5}}{2}}\times \operatorname{arsinh}\left(\frac{\sqrt{5}}{2}x+\frac{5}{2}\right)+c$$

$$= \frac{\sqrt{5}}{5}\operatorname{arsinh}\left(\frac{\sqrt{5}}{2}x+\frac{5}{2}\right)+c$$

36)

$$\int_{-1}^{1} \frac{1}{\sqrt{4x^2+8x+13}}\,\mathrm{d}x$$
$$= \int_{-1}^{1} \frac{1}{\sqrt{4(x^2+2x)+13}}\,\mathrm{d}x$$
$$= \int_{-1}^{1} \frac{1}{\sqrt{4((x+1)^2-1)+13}}\,\mathrm{d}x$$
$$= \int_{-1}^{1} \frac{1}{\sqrt{4(x+1)^2-4+13}}\,\mathrm{d}x$$
$$= \int_{-1}^{1} \frac{1}{\sqrt{4(x+1)^2+9}}\,\mathrm{d}x$$
$$= \int_{-1}^{1} \frac{1}{\sqrt{9\left(\frac{4}{9}(x+1)^2+1\right)}}\,\mathrm{d}x$$
$$= \frac{1}{3}\int_{-1}^{1} \frac{1}{\sqrt{\frac{4}{9}(x+1)^2+1}}\,\mathrm{d}x$$
$$= \frac{1}{3}\int_{-1}^{1} \frac{1}{\sqrt{\left(\frac{2}{3}x+\frac{2}{3}\right)^2+1}}\,\mathrm{d}x$$
$$= \frac{1}{3}\left[\frac{3}{2}\operatorname{arsinh}\left(\frac{2}{3}x+\frac{2}{3}\right)\right]_{-1}^{1}$$
$$= \frac{1}{2}\operatorname{arsinh}\frac{4}{3} - \frac{1}{2}\operatorname{arsinh} 0$$
$$= \frac{1}{2}\ln\left(\frac{4}{3}+\sqrt{\left(\frac{4}{3}\right)^2+1}\right) - 0$$
$$= \frac{1}{2}\ln\left(\frac{4}{3}+\frac{5}{3}\right)$$
$$= \frac{1}{2}\ln 3$$
$$\therefore k = 3$$

37) a)

$$-3-4x-x^2$$
$$\equiv -3-(x^2+4x)$$
$$\equiv -3-((x+2)^2-4)$$
$$\equiv -3-(x+2)^2+4$$
$$\equiv 1-(x+2)^2$$

37) b)

$-24 + 10x - x^2$

$\equiv -24 - (x^2 - 10x)$

$\equiv -24 - ((x-5)^2 - 25)$

$\equiv -24 - (x-5)^2 + 25$

$\equiv 1 - (x-5)^2$

37) c)

$-15 - 16x - 4x^2$

$\equiv -15 - 4(x^2 + 4x)$

$\equiv -15 - 4((x+2)^2 - 4)$

$\equiv -15 - 4(x+2)^2 + 16$

$\equiv 1 - 4(x+2)^2$

37) d)

$-143 + 72x - 9x^2$

$\equiv -143 - 9(x^2 - 8x)$

$\equiv -143 - 9((x-4)^2 - 16)$

$\equiv -143 - 9(x-4)^2 + 144$

$\equiv 1 - 9(x-4)^2$

37) e)

$-45 - 14x - x^2$

$\equiv -45 - (x^2 + 14x)$

$\equiv -45 - ((x+7)^2 - 49)$

$\equiv -45 - (x+7)^2 + 49$

$\equiv 4 - (x+7)^2$

37) f)

$8 + 4x - 4x^2$

$\equiv 8 - 4(x^2 - x)$

$\equiv 8 - 4\left(\left(x - \frac{1}{2}\right)^2 - \frac{1}{4}\right)$

$\equiv 8 - 4\left(x - \frac{1}{2}\right)^2 + 1$

$\equiv 9 - 4\left(x - \frac{1}{2}\right)^2$

38) a)

$-63 + 16x - x^2$

$\equiv -63 - (x^2 - 16x)$

$\equiv -63 - ((x-8)^2 - 64)$

$\equiv -63 - (x-8)^2 + 64$

$\equiv 1 - (x-8)^2$

38) b)

Let $x - 8 = \tanh u$

$\Rightarrow x = \tanh u + 8$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \operatorname{sech}^2 u$

$\Rightarrow \mathrm{d}x = \operatorname{sech}^2 u \ \mathrm{d}u$

$\therefore \int \frac{1}{-63+16x-x^2}\,\mathrm{d}x$

$= \int \frac{1}{1-(x-8)^2}\,\mathrm{d}x$

$= \int \frac{1}{1-\tanh^2 u} \times \operatorname{sech}^2 u \,\mathrm{d}u$

$= \int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\,\mathrm{d}u$

$= \int 1\,\mathrm{d}u$

$= u + c$

$= \operatorname{arctanh}(x-8) + c$

39) a)

$\int \frac{1}{-99-20x-x^2}\,\mathrm{d}x$

$= \int \frac{1}{-99-(x^2+20x)}\,\mathrm{d}x$

$= \int \frac{1}{-99-((x+10)^2-100)}\,\mathrm{d}x$

$= \int \frac{1}{-99-(x+10)^2+100}\,\mathrm{d}x$

$= \int \frac{1}{1-(x+10)^2}\,\mathrm{d}x$

$= \operatorname{artanh}(x+10) + c$

39) b)

$\int \frac{1}{-168+26x-x^2}\,\mathrm{d}x$

$= \int \frac{1}{-168-(x^2-26x)}\,\mathrm{d}x$

$= \int \frac{1}{-168-((x-13)^2-169)}\,\mathrm{d}x$

$= \int \frac{1}{-168-(x-13)^2+169}\,\mathrm{d}x$

$= \int \frac{1}{1-(x-13)^2}\,\mathrm{d}x$

$= \operatorname{artanh}(x+13) + c$

40) a)

$-35 + 24x - 4x^2$
$\equiv -35 - 4(x^2 - 6x)$
$\equiv -35 - 4((x-3)^2 - 9)$
$\equiv -35 - 4(x-3)^2 + 36$
$\equiv 1 - 4(x-3)^2$

40) b)

Let $2x - 6 = \tanh u$

$\Rightarrow x = \frac{1}{2}\tanh u + 3$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{2}\operatorname{sech}^2 u$

$\Rightarrow \mathrm{d}x = \frac{1}{2}\operatorname{sech}^2 u \ \mathrm{d}u$

$\therefore \int \frac{1}{-35+24x-4x^2}\,\mathrm{d}x$

$= \int \frac{1}{1-4(x-3)^2}\,\mathrm{d}x$

$= \int \frac{1}{1-(2x-6)^2}\,\mathrm{d}x$

$= \int \frac{1}{1-\tanh^2 u} \times \frac{1}{2}\operatorname{sech}^2 u\,\mathrm{d}u$

$= \frac{1}{2}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\,\mathrm{d}u$

$= \frac{1}{2}\int 1\,\mathrm{d}u$

$= \frac{1}{2}u + c$

$= \frac{1}{2}\operatorname{artanh}(2x - 6) + c$

41) a)

$\int \frac{1}{-224-90x-9x^2}\,\mathrm{d}x$

$= \int \frac{1}{-224-9(x^2+10x)}\,\mathrm{d}x$

$= \int \frac{1}{-224-9((x+5)^2-25)}\,\mathrm{d}x$

$= \int \frac{1}{-224-9(x+5)^2+225}\,\mathrm{d}x$

$= \int \frac{1}{1-9(x+5)^2}\,\mathrm{d}x$

$= \int \frac{1}{1-(3x+15)^2}\,\mathrm{d}x$

$= \frac{1}{3}\operatorname{artanh}(3x + 15) + c$

41) b)

$$\int \frac{1}{-24+50x-25x^2}\,\mathrm{d}x$$
$$= \int \frac{1}{-24-25(x^2-2x)}\,\mathrm{d}x$$
$$= \int \frac{1}{-24-25((x-1)^2-1)}\,\mathrm{d}x$$
$$= \int \frac{1}{-24-25(x-1)^2+25}\,\mathrm{d}x$$
$$= \int \frac{1}{1-25(x-1)^2}\,\mathrm{d}x$$
$$= \int \frac{1}{1-(5x-5)^2}\,\mathrm{d}x$$
$$= \frac{1}{5}\operatorname{artanh}(5x-5)+c$$

42) a)

$$11+10x-x^2$$
$$\equiv 11-(x^2-10x)$$
$$\equiv 11-((x-5)^2-25)$$
$$\equiv 11-(x-5)^2+25$$
$$\equiv 36-(x-5)^2$$

42) b)

Let $\frac{x-5}{6} = \tanh u$

$\Rightarrow x = 6\tanh u + 5$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 6\operatorname{sech}^2 u$

$\Rightarrow \mathrm{d}x = 6\operatorname{sech}^2 u \ \mathrm{d}u$

$\therefore \int \frac{1}{11+10x-x^2}\mathrm{d}x$

$= \int \frac{1}{36-(x-5)^2}\mathrm{d}x$

$= \int \frac{1}{36\left(1-\frac{1}{36}(x-5)^2\right)}\mathrm{d}x$

$= \frac{1}{36}\int \frac{1}{1-\left(\frac{1}{6}x-\frac{5}{6}\right)^2}\mathrm{d}x$

$= \frac{1}{36}\int \frac{1}{1-\tanh^2 u} \times 6\operatorname{sech}^2 u \,\mathrm{d}u$

$= \frac{1}{6}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\mathrm{d}u$

$= \frac{1}{6}\int 1 \,\mathrm{d}u$

$= \frac{1}{6}u + c$

$= \frac{1}{6}\operatorname{arctanh}\left(\frac{1}{6}x - \frac{5}{6}\right) + c$

43) a)

$\int \frac{1}{91-18x-9x^2}\mathrm{d}x$

$= \int \frac{1}{91-9(x^2+2x)}\mathrm{d}x$

$= \int \frac{1}{91-9((x+1)^2-1)}\mathrm{d}x$

$= \int \frac{1}{91-9(x+1)^2+9}\mathrm{d}x$

$= \int \frac{1}{100-9(x+1)^2}\mathrm{d}x$

$= \int \frac{1}{100\left(1-\frac{9}{100}(x+1)^2\right)}\mathrm{d}x$

$= \frac{1}{100}\int \frac{1}{1-\left(\frac{3}{10}x+\frac{3}{10}\right)^2}\mathrm{d}x$

$= \frac{1}{100} \times \frac{1}{\frac{3}{10}} \times \operatorname{artanh}\left(\frac{3}{10}x + \frac{3}{10}\right) + c$

$= \frac{1}{30}\operatorname{artanh}\left(\frac{3}{10}x + \frac{3}{10}\right) + c$

43) b)

$$\int \frac{1}{-20-24x-4x^2}\,\mathrm{d}x$$
$$= \int \frac{1}{-20-4(x^2+6x)}\,\mathrm{d}x$$
$$= \int \frac{1}{-20-4((x+3)^2-9)}\,\mathrm{d}x$$
$$= \int \frac{1}{-20-4(x+3)^2+36}\,\mathrm{d}x$$
$$= \int \frac{1}{16-4(x+3)^2}\,\mathrm{d}x$$
$$= \int \frac{1}{16\left(1-\frac{1}{4}(x+3)^2\right)}\,\mathrm{d}x$$
$$= \frac{1}{16}\int \frac{1}{1-\left(\frac{1}{2}x+\frac{3}{2}\right)^2}\,\mathrm{d}x$$
$$= \frac{1}{16} \times \frac{1}{\frac{1}{2}} \times \operatorname{artanh}\left(\frac{1}{2}x+\frac{3}{2}\right)+c$$
$$= \frac{1}{8}\operatorname{artanh}\left(\frac{1}{2}x+\frac{3}{2}\right)+c$$

44)

$$\int_0^5 \frac{5}{63+12x-3x^2}\,\mathrm{d}x$$
$$= \int_0^5 \frac{5}{63-3(x^2-4x)}\,\mathrm{d}x$$
$$= \int_0^5 \frac{5}{63-3((x-2)^2-4)}\,\mathrm{d}x$$
$$= \int_0^5 \frac{5}{63-3(x-2)^2+12}\,\mathrm{d}x$$
$$= \int_0^5 \frac{5}{75-3(x-2)^2}\,\mathrm{d}x$$
$$= \int_0^5 \frac{5}{75\left(1-\frac{1}{25}(x-2)^2\right)}\,\mathrm{d}x$$
$$= \frac{1}{15}\int_0^5 \frac{1}{1-\left(\frac{1}{5}x-\frac{2}{5}\right)^2}\,\mathrm{d}x$$
$$= \frac{1}{15}\left[\frac{1}{\frac{1}{5}} \times \operatorname{artanh}\left(\frac{1}{5}x-\frac{2}{5}\right)\right]_0^5$$
$$= \left[\frac{1}{3}\operatorname{artanh}\left(\frac{1}{5}x-\frac{2}{5}\right)\right]_0^5$$
$$= \frac{1}{3}\operatorname{artanh}\frac{3}{5} - \frac{1}{3}\operatorname{artanh}\left(-\frac{2}{5}\right)$$
$$= \frac{1}{3}\times\frac{1}{2}\ln\left(\frac{1+\frac{3}{5}}{1-\frac{3}{5}}\right) - \frac{1}{3}\times\frac{1}{2}\ln\left(\frac{1+\left(-\frac{2}{5}\right)}{1-\left(-\frac{2}{5}\right)}\right)$$
$$= \frac{1}{6}\ln 4 - \frac{1}{6}\ln\frac{3}{7}$$
$$= \frac{1}{6}\left(\ln 4 - \ln\frac{3}{7}\right)$$
$$= \frac{1}{6}\ln\frac{28}{3}$$

45)

Volume

$$= \pi \int_2^3 \left(\frac{2}{\sqrt{-16+16x-3x^2}} \right)^2 \mathrm{d}x$$

$$= \pi \int_2^3 \frac{4}{-16+16x-3x^2} \mathrm{d}x$$

$$= 4\pi \int_2^3 \frac{1}{-16-3\left(x^2-\frac{16}{3}x\right)} \mathrm{d}x$$

$$= 4\pi \int_2^3 \frac{1}{-16-3\left(\left(x-\frac{8}{3}\right)^2-\frac{64}{9}\right)} \mathrm{d}x$$

$$= 4\pi \int_2^3 \frac{1}{-16-3\left(x-\frac{8}{3}\right)^2+\frac{64}{3}} \mathrm{d}x$$

$$= 4\pi \int_2^3 \frac{1}{\frac{16}{3}-3\left(x-\frac{8}{3}\right)^2} \mathrm{d}x$$

$$= 4\pi \int_2^3 \frac{1}{\frac{16}{3}\left(1-\frac{9}{16}\left(x-\frac{8}{3}\right)^2\right)} \mathrm{d}x$$

$$= \frac{3\pi}{4} \int_2^3 \frac{1}{1-\left(\frac{3}{4}x-2\right)^2} \mathrm{d}x$$

$$= \frac{3\pi}{4} \left[\frac{4}{3} \operatorname{artanh}\left(\frac{3}{4}x-2\right) \right]_2^3$$

$$= \frac{3\pi}{4} \left(\frac{4}{3} \operatorname{artanh} \frac{1}{4} - \frac{4}{3} \operatorname{artanh}\left(-\frac{1}{2}\right) \right)$$

$$= \frac{3\pi}{4} \left(\frac{4}{3} \times \frac{1}{2} \ln\left(\frac{1+\frac{1}{4}}{1-\frac{1}{4}} \right) - \frac{4}{3} \times \frac{1}{2} \ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)} \right) \right)$$

$$= \frac{3\pi}{4} \left(\frac{2}{3} \ln \frac{5}{3} - \frac{2}{3} \ln \frac{1}{3} \right)$$

$$= \frac{\pi}{2} \left(\ln \frac{5}{3} - \ln \frac{1}{3} \right)$$

$$= \frac{\pi}{2} \ln 5 \text{ units}^3$$

46)

$\int \sqrt{x^2-1}\,\mathrm{d}x$

Let $x = \cosh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sinh u \Rightarrow \mathrm{d}x = \sinh u\,\mathrm{d}u$

$\therefore \int \sqrt{x^2-1}\,\mathrm{d}x$

$= \int \sqrt{\cosh^2 u - 1} \times \sinh u\,\mathrm{d}u$

$= \int \sqrt{\sinh^2 u} \times \sinh u\,\mathrm{d}u$

$= \int \sinh u \times \sinh u\,\mathrm{d}u$

$= \int \sinh^2 u\,\mathrm{d}u$

$\cosh 2u \equiv 1 + 2\sinh^2 u \Rightarrow \sinh^2 u \equiv \frac{1}{2}(\cosh 2u - 1)$

$\therefore \int \sinh^2 u\,\mathrm{d}u = \int \left(\frac{1}{2}(\cosh 2u - 1)\right)\mathrm{d}u$

$= \frac{1}{2}\int (\cosh 2u - 1)\,\mathrm{d}u$

$= \frac{1}{2}\left(\frac{1}{2}\sinh 2u - u\right) + c$

$= \frac{1}{4}\sinh 2u - \frac{1}{2}u + c$

$\sinh 2u = 2\sinh u \cosh u = 2\sqrt{\cosh^2 u - 1} \times \cosh u = 2\sqrt{x^2-1} \times x = 2x\sqrt{x^2-1}$

$\therefore \int \sqrt{x^2-1}\,\mathrm{d}x = \frac{1}{4} \times 2x\sqrt{x^2-1} - \frac{1}{2}\operatorname{arcosh} x + c$

$= \frac{1}{2}x\sqrt{x^2-1} - \frac{1}{2}\operatorname{arcosh} x + c$

47)

$\int \sqrt{x^2+1}\,\mathrm{d}x$

Let $x = \sinh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u\,\mathrm{d}u$

$\therefore \int \sqrt{x^2+1}\,\mathrm{d}x$

$= \int \sqrt{\sinh^2 u - 1} \times \cosh u\,\mathrm{d}u$

$= \int \sqrt{\cosh^2 u} \times \cosh u\,\mathrm{d}u$

$= \int \cosh u \times \cosh u\,\mathrm{d}u$

$= \int \cosh^2 u\,\mathrm{d}u$

$\cosh 2u \equiv 2\cosh^2 u - 1 \Rightarrow \cosh^2 u \equiv \frac{1}{2}(\cosh 2u + 1)$

$\therefore \int \cosh^2 u\,\mathrm{d}u = \int \left(\frac{1}{2}(\cosh 2u + 1)\right)\mathrm{d}u$

$= \frac{1}{2}\int (\cosh 2u + 1)\,\mathrm{d}u$

$= \frac{1}{2}\left(\frac{1}{2}\sinh 2u + u\right) + c$

$= \frac{1}{4}\sinh 2u + \frac{1}{2}u + c$

$\sinh 2u = 2\sinh u \cosh u = 2\sinh u \times \sqrt{\sinh^2 u + 1} = 2x\sqrt{x^2+1}$

$\therefore \int \sqrt{x^2+1}\,\mathrm{d}x = \frac{1}{4} \times 2x\sqrt{x^2+1} + \frac{1}{2}\operatorname{arsinh} x + c$

$= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\operatorname{arsinh} x + c$

48)

$\int \sqrt{9x^2-1}\,\mathrm{d}x$

Let $x = \frac{1}{3}\cosh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{3}\sinh u \Rightarrow \mathrm{d}x = \frac{1}{3}\sinh u\,\mathrm{d}u$

$\therefore \int \sqrt{9x^2-1}\,\mathrm{d}x$

$= \int \sqrt{9\left(\frac{1}{3}\cosh u\right)^2 - 1} \times \frac{1}{3}\sinh u\,\mathrm{d}u$

$= \frac{1}{3}\int \sqrt{\cosh^2 u - 1} \times \sinh u\,\mathrm{d}u$

$= \frac{1}{3}\int \sqrt{\sinh^2 u} \times \sinh u\,\mathrm{d}u$

$= \frac{1}{3}\int \sinh u \times \sinh u\,\mathrm{d}u$

$= \frac{1}{3}\int \sinh^2 u\,\mathrm{d}u$

$\cosh 2u = 1 + 2\sinh^2 u \Rightarrow \sinh^2 u = \frac{1}{2}(\cosh 2u - 1)$

$\therefore \frac{1}{3}\int \sinh^2 u\,\mathrm{d}u$

$= \frac{1}{3}\int \left(\frac{1}{2}(\cosh 2u - 1)\right)\mathrm{d}u$

$= \frac{1}{6}\int (\cosh 2u - 1)\,\mathrm{d}u$

$= \frac{1}{6}\left(\frac{1}{2}\sinh 2u - u\right) + c$

$= \frac{1}{12}\sinh 2u - \frac{1}{6}u + c$

$x = \frac{1}{3}\cosh u \Rightarrow \cosh u = 3x$

$\sinh 2u = 2\sinh u \cosh u = 2\sqrt{\cosh^2 u - 1} \times \cosh u = 2\sqrt{(3x)^2 - 1} \times 3x = 6x\sqrt{9x^2-1}$

$\therefore \int \sqrt{9x^2-1}\,\mathrm{d}x$

$= \frac{1}{12} \times 6x\sqrt{9x^2-1} - \frac{1}{6}\operatorname{arcosh} 3x + c$

$= \frac{1}{2}x\sqrt{9x^2-1} - \frac{1}{6}\operatorname{arcosh} 3x + c$

49)

$\int_{12}^{13} \sqrt{x^2 - 144}\,\mathrm{d}x$

Let $x = 12\cosh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 12\sinh u \Rightarrow dx = 12\sinh u\,\mathrm{d}u$

Change limits:

$x = 13 \Rightarrow 13 = 12\cosh u \Rightarrow \cosh u = \frac{13}{12} \Rightarrow u = \operatorname{arcosh}\frac{13}{12} = \ln\left(\frac{13}{12} + \sqrt{\left(\frac{13}{12}\right)^2 - 1}\right)$

$= \ln\left(\frac{13}{12} + \frac{5}{12}\right) = \ln\frac{3}{2}$

$x = 12 \Rightarrow 12 = 12\cosh u \Rightarrow \cosh u = 1 \Rightarrow u = 0$

$\therefore \int_{12}^{13} \sqrt{x^2 - 144}\,\mathrm{d}x = \int_0^{\ln\frac{3}{2}} \sqrt{(12\cosh u)^2 - 144} \times 12\sinh u\,\mathrm{d}u$

$= \int_0^{\ln\frac{3}{2}} \sqrt{144\cosh^2 u - 144} \times 12\sinh u\,\mathrm{d}u$

$= \int_0^{\ln\frac{3}{2}} 12\sqrt{\cosh^2 u - 1} \times 12\sinh u\,\mathrm{d}u$

$= 144\int_0^{\ln\frac{3}{2}} \sqrt{\sinh^2 u} \times \sinh u\,\mathrm{d}u$

$= 144\int_0^{\ln\frac{3}{2}} \sinh^2 u\,\mathrm{d}u$

$\cosh 2u \equiv 1 + 2\sinh^2 u \Rightarrow \sinh^2 u \equiv \frac{1}{2}(\cosh 2u - 1)$

$\therefore 144\int_0^{\ln\frac{3}{2}} \sinh^2 u\,\mathrm{d}u = 144\int_0^{\ln\frac{3}{2}} \frac{1}{2}(\cosh 2u - 1)\,\mathrm{d}u$

$= 72\int_0^{\ln\frac{3}{2}} (\cosh 2u - 1)\,\mathrm{d}u$

$= 72\left[\frac{1}{2}\sinh 2u - u\right]_0^{\ln\frac{3}{2}}$

$= 72\left(\left(\frac{1}{2}\sinh\left(2\ln\frac{3}{2}\right) - \ln\frac{3}{2}\right) - \left(\frac{1}{2}\sinh 0 - 0\right)\right)$

$= 36 \times \frac{\mathrm{e}^{2\ln\frac{3}{2}} - \mathrm{e}^{-2\ln\frac{3}{2}}}{2} - 72\ln\frac{3}{2}$

$= 18\left(\mathrm{e}^{\ln\left(\frac{3}{2}\right)^2} - \mathrm{e}^{\ln\left(\frac{3}{2}\right)^{-2}}\right) - 72\ln\frac{3}{2}$

$= 18\left(\mathrm{e}^{\ln\frac{9}{4}} - \mathrm{e}^{\ln\frac{4}{9}}\right) - 72\ln\frac{3}{2}$

$= 18\left(\frac{9}{4} - \frac{4}{9}\right) - 72\ln\frac{3}{2}$

$= \frac{65}{2} - 72\ln\frac{3}{2}$

50)

$\int_0^2 \sqrt{25x^2+1}\,\mathrm{d}x$

Let $x = \frac{1}{5}\sinh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{5}\cosh u \Rightarrow \mathrm{d}x = \frac{1}{5}\cosh u\,\mathrm{d}u$

Change limits:

$x = 2 \Rightarrow 2 = \frac{1}{5}\sinh u \Rightarrow \sinh u = 10 \Rightarrow u = \ln\left(10+\sqrt{10^2+1}\right) = \ln\left(10+\sqrt{101}\right)$

$x = 0 \Rightarrow 0 = \frac{1}{5}\sinh u \Rightarrow \sinh u = 0 \Rightarrow u = 0$

$\therefore \int_0^2 \sqrt{25x^2+1}\,\mathrm{d}x = \int_0^{\ln\left(10+\sqrt{101}\right)} \sqrt{25\left(\frac{1}{5}\sinh u\right)^2+1} \times \frac{1}{5}\cosh u\,\mathrm{d}u$

$= \frac{1}{5}\int_0^{\ln\left(10+\sqrt{101}\right)} \sqrt{\sinh^2 u+1} \times \cosh u\,\mathrm{d}u$

$= \frac{1}{5}\int_0^{\ln\left(10+\sqrt{101}\right)} \sqrt{\cosh^2 u} \times \cosh u\,\mathrm{d}u$

$= \frac{1}{5}\int_0^{\ln\left(10+\sqrt{101}\right)} \cosh^2 u\,\mathrm{d}u$

$\cosh 2u \equiv 2\cosh^2 u - 1 \Rightarrow \cosh^2 u \equiv \frac{1}{2}(\cosh 2u + 1)$

$\therefore \frac{1}{5}\int_0^{\ln\left(10+\sqrt{101}\right)} \cosh^2 u\,\mathrm{d}u = \frac{1}{5}\int_0^{\ln\left(10+\sqrt{101}\right)} \left(\frac{1}{2}(\cosh 2u+1)\right)\mathrm{d}u$

$= \frac{1}{10}\int_0^{\ln\left(10+\sqrt{101}\right)} (\cosh 2u+1)\,\mathrm{d}u$

$= \frac{1}{10}\left[\frac{1}{2}\sinh 2u + u\right]_0^{\ln\left(10+\sqrt{101}\right)}$

$= \frac{1}{10}\left(\left(\frac{1}{2}\sinh(2\ln\left(10+\sqrt{101}\right) + \ln\left(10+\sqrt{101}\right)\right) - \left(\frac{1}{2}\sinh 0 + 0\right)\right)$

$= \frac{1}{10}\left(\left(\frac{1}{2} \times \frac{\mathrm{e}^{2\ln\left(10+\sqrt{101}\right)} - \mathrm{e}^{-2\ln\left(10+\sqrt{101}\right)}}{2} + \ln\left(10+\sqrt{101}\right)\right) - 0\right)$

$= \frac{1}{40}\left(\mathrm{e}^{\ln\left(10+\sqrt{101}\right)^2} - \mathrm{e}^{\ln\left(10+\sqrt{101}\right)^{-2}}\right) + \frac{1}{10}\ln\left(10+\sqrt{101}\right)$

$= \frac{1}{40}\left(\mathrm{e}^{\ln\left(201+20\sqrt{101}\right)} - \mathrm{e}^{\ln\left(201-20\sqrt{101}\right)}\right) + \frac{1}{10}\ln\left(10+\sqrt{101}\right)$

$= \frac{1}{40}\left(201+20\sqrt{101} - \left(201-20\sqrt{101}\right)\right) + \frac{1}{10}\ln\left(10+\sqrt{101}\right)$

$= \frac{1}{40}\left(40\sqrt{101}\right) + \frac{1}{10}\ln\left(10+\sqrt{101}\right)$

$= \sqrt{101} + \frac{1}{10}\ln\left(10+\sqrt{101}\right)$

51) a)

$\int \sqrt{x^2 - 10x + 26}\,\mathrm{d}x = \int \sqrt{(x-5)^2 - 25 + 26}\,\mathrm{d}x$

$= \int \sqrt{(x-5)^2 + 1}\,\mathrm{d}x$

Let $x - 5 = \sinh u \Rightarrow x = \sinh u + 5 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u\,\mathrm{d}u$

$\therefore \int \sqrt{(x-5)^2 + 1}\,\mathrm{d}x = \int \sqrt{\sinh^2 u + 1} \times \cosh u\,\mathrm{d}u$

$= \int \sqrt{\cosh^2 u} \times \cosh u\,\mathrm{d}u$

$= \int \cosh^2 u\,\mathrm{d}u$

$\cosh 2u \equiv 2\cosh^2 u - 1 \Rightarrow \cosh^2 u \equiv \frac{1}{2}(\cosh 2u + 1)$

$\therefore \int \cosh^2 u\,\mathrm{d}u = \int \frac{1}{2}(\cosh 2u + 1)\,\mathrm{d}u$

$= \frac{1}{2}\int (\cosh 2u + 1)\,\mathrm{d}u$

$= \frac{1}{2}\left(\frac{1}{2}\sinh 2u + u\right) + c$

$= \frac{1}{4}\sinh 2u + \frac{1}{2}u + c$

$\sinh 2u \equiv 2\sinh u \cosh u \equiv 2\sinh u\sqrt{\sinh^2 u + 1}$

$\therefore \frac{1}{4}\sinh 2u + \frac{1}{2}u + c = \frac{1}{4} \times 2\sinh u\sqrt{\sinh^2 u + 1} + \frac{1}{2}u + c$

$= \frac{1}{2}(x-5)\sqrt{(x-5)^2 + 1} + \frac{1}{2}\operatorname{arsinh}(x-5) + c$

$= \frac{1}{2}(x-5)\sqrt{(x-5)^2 + 1} + \frac{1}{2}\ln\left(x - 5 + \sqrt{(x-5)^2 + 1}\right) + c$

$\therefore \frac{1}{2}(x-5)\sqrt{(x-5)^2 + 1} + \frac{1}{2}\operatorname{arsinh}(x-5) + c$ or $\frac{1}{2}(x-5)\sqrt{(x-5)^2 + 1} +$

$\frac{1}{2}\ln\left(x - 5 + \sqrt{(x-5)^2 + 1}\right) + c$

51) b)

$\int \sqrt{x^2+6x+8}\,\mathrm{d}x = \int \sqrt{(x+3)^2-9+8}\,\mathrm{d}x$

$= \int \sqrt{(x+3)^2-1}\,\mathrm{d}x$

Let $x+3 = \cosh u \Rightarrow x = \cosh u - 3 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sinh u \Rightarrow \mathrm{d}x = \sinh u\,\mathrm{d}u$

$\therefore \int \sqrt{(x+3)^2-1}\,\mathrm{d}x = \int \sqrt{\cosh^2 u - 1} \times \sinh u\,\mathrm{d}u$

$= \int \sqrt{\sinh^2 u} \times \sinh u\,\mathrm{d}u$

$= \int \sinh^2 u\,\mathrm{d}u$

$\cosh 2u \equiv 1 + 2\sinh^2 u \Rightarrow \sinh^2 u \equiv \frac{1}{2}(\cosh 2u - 1)$

$\therefore \int \sinh^2 u\,\mathrm{d}u = \int \frac{1}{2}(\cosh 2u - 1)\,\mathrm{d}u$

$= \frac{1}{2}\int (\cosh 2u - 1)\,\mathrm{d}u$

$= \frac{1}{2}\left(\frac{1}{2}\sinh 2u - u\right) + c$

$= \frac{1}{4}\sinh 2u - \frac{1}{2}u + c$

$\sinh 2u \equiv 2\sinh u \cosh u \equiv 2\sqrt{\cosh^2 u - 1}\cosh u$

$\therefore \frac{1}{4}\sinh 2u - \frac{1}{2}u + c = \frac{1}{4} \times 2\sqrt{\cosh^2 u - 1}\cosh u - \frac{1}{2}u + c$

$= \frac{1}{2}(x+3)\sqrt{(x+3)^2-1} - \frac{1}{2}\operatorname{arcosh}(x+3) + c$

$= \frac{1}{2}(x+3)\sqrt{(x+3)^2-1} - \frac{1}{2}\ln\left(x+3+\sqrt{(x+3)^2-1}\right) + c$

$\therefore \frac{1}{2}(x+3)\sqrt{(x+3)^2-1} - \frac{1}{2}\operatorname{arcosh}(x+3) + c$ or $\frac{1}{2}(x+3)\sqrt{(x+3)^2-1} - \frac{1}{2}\ln\left(x+3+\sqrt{(x+3)^2-1}\right) + c$

52)

$\int_3^5 \sqrt{x^2-8x+17}\,\mathrm{d}x = \int_3^5 \sqrt{(x-4)^2-16+17}\,\mathrm{d}x = \int_3^5 \sqrt{(x-4)^2+1}\,\mathrm{d}x$

Let $x-4 = \sinh u \Rightarrow x = \sinh u + 4 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u\,\mathrm{d}u$

Change limits:

$x = 5 \Rightarrow 1 = \sinh u \Rightarrow u = \ln\left(1+\sqrt{1^2+1}\right) = \ln\left(1+\sqrt{2}\right)$

$x = 3 \Rightarrow -1 = \sinh u \Rightarrow u = \ln\left(-1+\sqrt{(-1)^2+1}\right) = \ln\left(-1+\sqrt{2}\right)$

$\therefore \int_3^5 \sqrt{(x-4)^2+1}\,\mathrm{d}x = \int_{\ln(-1+\sqrt{2})}^{\ln(1+\sqrt{2})} \sqrt{\sinh^2 u + 1} \times \cosh u\,\mathrm{d}u$

$= \int_{\ln(-1+\sqrt{2})}^{\ln(1+\sqrt{2})} \sqrt{\cosh^2 u} \times \cosh u\,\mathrm{d}u$

$= \int_{\ln(-1+\sqrt{2})}^{\ln(1+\sqrt{2})} \cosh^2 u\,\mathrm{d}u$

$\cosh 2u \equiv 2\cosh^2 u - 1 \Rightarrow \cosh^2 u \equiv \frac{1}{2}(\cosh 2u + 1)$

$\therefore \int_{\ln(-1+\sqrt{2})}^{\ln(1+\sqrt{2})} \cosh^2 u \, du = \int_{\ln(-1+\sqrt{2})}^{\ln(1+\sqrt{2})} \left(\frac{1}{2}(\cosh 2u + 1)\right) du$

$= \frac{1}{2}\int_{\ln(-1+\sqrt{2})}^{\ln(1+\sqrt{2})} (\cosh 2u + 1)\, du$

$= \frac{1}{2}\left[\frac{1}{2}\sinh 2u + u\right]_{\ln(-1+\sqrt{2})}^{\ln(1+\sqrt{2})}$

$= \frac{1}{2}\left(\left(\frac{1}{2}\sinh\left(2\ln\left(1+\sqrt{2}\right)\right) + \ln\left(1+\sqrt{2}\right)\right) - \left(\frac{1}{2}\sinh\left(2\ln\left(-1+\sqrt{2}\right)\right) + \ln\left(-1+\sqrt{2}\right)\right)\right)$

$= \frac{1}{2}\left(\left(\frac{1}{2} \times \frac{e^{2\ln(1+\sqrt{2})} - e^{-2\ln(1+\sqrt{2})}}{2} + \ln\left(1+\sqrt{2}\right)\right) - \left(\frac{1}{2} \times \frac{e^{2\ln(-1+\sqrt{2})} - e^{-2\ln(-1+\sqrt{2})}}{2} + \ln\left(-1+\sqrt{2}\right)\right)\right)$

$= \frac{1}{2}\left(\left(\frac{1}{4}\left(e^{\ln(1+\sqrt{2})^2} - e^{\ln(1+\sqrt{2})^{-2}}\right) + \ln\left(1+\sqrt{2}\right)\right) - \left(\frac{1}{4}\left(e^{\ln(-1+\sqrt{2})^2} - e^{\ln(-1+\sqrt{2})^{-2}}\right) + \ln\left(-1+\sqrt{2}\right)\right)\right)$

$= \frac{1}{2}\left(\left(\frac{1}{4}\left(e^{\ln(3+2\sqrt{2})} - e^{\ln(3-2\sqrt{2})}\right) + \ln\left(1+\sqrt{2}\right)\right) - \left(\frac{1}{4}\left(e^{\ln(3-2\sqrt{2})} - e^{\ln(3+2\sqrt{2})}\right) + \ln\left(-1+\sqrt{2}\right)\right)\right)$

$= \frac{1}{2}\left(\left(\frac{1}{4}\left(3+2\sqrt{2} - (3-2\sqrt{2})\right) + \ln\left(1+\sqrt{2}\right)\right) - \left(\frac{1}{4}\left(3-2\sqrt{2} - (3+2\sqrt{2})\right) + \ln\left(-1+\sqrt{2}\right)\right)\right)$

$= \frac{1}{2}\left(\left(\sqrt{2} + \ln\left(1+\sqrt{2}\right)\right) - \left(-\sqrt{2} + \ln\left(-1+\sqrt{2}\right)\right)\right)$

$= \frac{1}{2}\left(2\sqrt{2} + \ln\left(1+\sqrt{2}\right) - \ln\left(-1+\sqrt{2}\right)\right)$

$= \sqrt{2} + \frac{1}{2}\ln\left(\frac{1+\sqrt{2}}{-1+\sqrt{2}}\right)$

$= \sqrt{2} + \frac{1}{2}\ln\left(3+2\sqrt{2}\right)$

$= \sqrt{2} + \ln\left(3+2\sqrt{2}\right)^{\frac{1}{2}}$

$= \sqrt{2} + \ln\left(1+\sqrt{2}\right)$

53)

$\int \frac{x^2}{\sqrt{x^2-1}}\mathrm{d}x$

Let $x = \cosh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sinh u \Rightarrow \mathrm{d}x = \sinh u\,\mathrm{d}u$

$\therefore \int \frac{x^2}{\sqrt{x^2-1}}\mathrm{d}x$

$= \int \frac{\cosh^2 u}{\sqrt{\cosh^2 u-1}} \times \sinh u\,\mathrm{d}u$

$= \int \frac{\cosh^2 u}{\sqrt{\sinh^2 u}} \times \sinh u\,\mathrm{d}u$

$= \int \frac{\cosh^2 u}{\sinh u} \times \sinh u\,\mathrm{d}u$

$= \int \cosh^2 u\,\mathrm{d}u$

$\cosh 2u = 2\cosh^2 u - 1 \Rightarrow \cosh^2 u = \frac{1}{2}(\cosh 2u + 1)$

$\therefore \int \cosh^2 u\,\mathrm{d}u$

$= \int \left(\frac{1}{2}(\cosh 2u + 1)\right)\mathrm{d}u$

$= \frac{1}{2}\int (\cosh 2u + 1)\,\mathrm{d}u$

$= \frac{1}{2}\left(\frac{1}{2}\sinh 2u + u\right) + c$

$= \frac{1}{4}\sinh 2u + \frac{1}{2}u + c$

$\sinh 2u = 2\sinh u \cosh u = 2\sqrt{\cosh^2 u - 1} \times \cosh u = 2\sqrt{x^2-1} \times x = 2x\sqrt{x^2-1}$

$\therefore \int \frac{x^2}{\sqrt{x^2-1}}\mathrm{d}x = \frac{1}{4} \times 2x\sqrt{x^2-1} + \frac{1}{2}\operatorname{arcosh} x + c$

$= \frac{1}{2}x\sqrt{x^2-1} + \frac{1}{2}\operatorname{arcosh} x + c$

$\therefore \frac{1}{2}x\sqrt{x^2-1} + \frac{1}{2}\operatorname{arcosh} x + c$ or $\frac{1}{2}x\sqrt{x^2-1} + \frac{1}{2}\ln\left(x + \sqrt{x^2-1}\right) + c$

54)

$\int \frac{x^2}{\sqrt{x^2+1}}\mathrm{d}x$

Let $x = \sinh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u\,\mathrm{d}u$

$\therefore \int \frac{x^2}{\sqrt{x^2+1}}\mathrm{d}x = \int \frac{\sinh^2 u}{\sqrt{\sinh^2 u+1}} \times \cosh u\,\mathrm{d}u$

$= \int \frac{\sinh^2 u}{\sqrt{\cosh^2 u}} \times \cosh u\,\mathrm{d}u$

$= \int \frac{\sinh^2 u}{\cosh u} \times \cosh u\,\mathrm{d}u$

$= \int \sinh^2 u\,\mathrm{d}u$

$\cosh 2u = 1 + 2\sinh^2 u \Rightarrow \sinh^2 u = \frac{1}{2}(\cosh 2u - 1)$

$\therefore \int \sinh^2 u\,\mathrm{d}u = \int \left(\frac{1}{2}(\cosh 2u - 1)\right)\mathrm{d}u$

$= \frac{1}{2}\int (\cosh 2u - 1)\,\mathrm{d}u$

$= \frac{1}{2}\left(\frac{1}{2}\sinh 2u - u\right) + c$

$= \frac{1}{4}\sinh 2u - \frac{1}{2}u + c$

$\sinh 2u = 2\sinh u \cosh u = 2\sinh u \times \sqrt{\sinh^2 u + 1} = 2 \times x \times \sqrt{x^2+1} = 2x\sqrt{x^2+1}$

$\therefore \int \frac{x^2}{\sqrt{x^2+1}}\mathrm{d}x = \frac{1}{4} \times 2x\sqrt{x^2+1} - \frac{1}{2}\mathrm{arsinh}\, x + c$

$= \frac{1}{2}x\sqrt{x^2+1} - \frac{1}{2}\mathrm{arsinh}\, x + c$

$\therefore \frac{1}{2}x\sqrt{x^2+1} - \frac{1}{2}\mathrm{arsinh}\, x + c$ or $\frac{1}{2}x\sqrt{x^2+1} - \frac{1}{2}\ln\left(x + \sqrt{x^2+1}\right) + c$

55)

$\int \frac{x^3}{\sqrt{x^2+1}}\,\mathrm{d}x$

Let $x = \sinh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u\,\mathrm{d}u$

$\therefore \int \frac{x^3}{\sqrt{x^2+1}}\mathrm{d}x = \int \frac{\sinh^3 u}{\sqrt{\sinh^2 u+1}} \times \cosh u\,\mathrm{d}u$

$= \int \frac{\sinh^3 u}{\sqrt{\cosh^2 u}} \times \cosh u\,\mathrm{d}u$

$= \int \frac{\sinh^3 u}{\cosh u} \times \cosh u\,\mathrm{d}u$

$= \int \sinh^3 u\,\mathrm{d}u$

$= \int \sinh^2 u \times \sinh u\,\mathrm{d}u$

$= \int (\cosh^2 u - 1)\sinh u\,\mathrm{d}u$

$= \int (\cosh^2 u \sinh u - \sinh u)\,\mathrm{d}u$

$= \frac{1}{3}\cosh^3 u - \cosh u + c$

$x = \sinh u \Rightarrow x^2 = \sinh^2 u \Rightarrow x^2 + 1 = \sinh^2 u + 1 \Rightarrow x^2 + 1 = \cosh^2 u$

$\therefore \int \frac{x^3}{\sqrt{x^2+1}}\mathrm{d}x = \frac{1}{3}(x^2+1)^{\frac{3}{2}} - \sqrt{x^2+1} + c$

56)

$\int \sqrt{\frac{x}{x+1}}\,\mathrm{d}x$

Use Integration by Substitution:

Let $x = \sinh^2 u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\sinh u \cosh u \Rightarrow \mathrm{d}x = 2\sinh u \cosh u\,\mathrm{d}u$

$\therefore \int \sqrt{\frac{x}{x+1}}\,\mathrm{d}x = \int \sqrt{\frac{\sinh^2 u}{\sinh^2 u+1}} \times 2\sinh u \cosh u\,\mathrm{d}u$

$= \int \sqrt{\frac{\sinh^2 u}{\cosh^2 u}} \times 2\sinh u \cosh u\,\mathrm{d}u$

$= \int \frac{\sinh u}{\cosh u} \times 2\sinh u \cosh u\,\mathrm{d}u$

$= \int 2\sinh^2 u\,\mathrm{d}u$

$= \int (\cosh 2u - 1)\,\mathrm{d}u$

$= \frac{1}{2}\sinh 2u - u + c$

$= \sinh u \cosh u - u + c$

$= \sqrt{\sinh^2 u}\sqrt{\cosh^2 u} - u + c$

$= \sqrt{\sinh^2 u}\sqrt{\sinh^2 u + 1} - u + c$

$= \sqrt{x}\sqrt{x+1} - \operatorname{arsinh}\left(\sqrt{x}\right) + c$

$= \sqrt{x^2 + x} - \operatorname{arsinh}\left(\sqrt{x}\right) + c$

$\therefore \sqrt{x^2 + x} - \operatorname{arsinh}\left(\sqrt{x}\right) + c$ or $\sqrt{x^2 + x} - \ln\left(\sqrt{x} + \sqrt{x+1}\right) + c$

57)

$\int (x^2-1)^{\frac{3}{2}}\,\mathrm{d}x$

Let $x = \cosh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sinh u \Rightarrow \mathrm{d}x = \sinh u\,\mathrm{d}u$

$\therefore \int (x^2-1)^{\frac{3}{2}}\,\mathrm{d}x$

$= \int (\cosh^2 u - 1)^{\frac{3}{2}} \times \sinh u\,\mathrm{d}u$

$= \int (\sinh^2 u)^{\frac{3}{2}} \times \sinh u\,\mathrm{d}u$

$= \int \sinh^3 u \times \sinh u\,\mathrm{d}u$

$= \int \sinh^4 u\,\mathrm{d}u$

$\cosh 2A \equiv 1 + 2\sinh^2 A \Rightarrow \sinh^2 A \equiv \frac{1}{2}(\cosh 2A - 1)$

$\therefore \int \sinh^4 u\,\mathrm{d}u = \int (\sinh^2 u)^2\,\mathrm{d}u = \int \left(\left(\frac{1}{2}(\cosh 2u - 1)\right)^2\right)\mathrm{d}u$

$= \frac{1}{4}\int (\cosh^2 2u - 2\cosh 2u + 1)\,\mathrm{d}u$

$\cosh 2A \equiv 2\cosh^2 A - 1 \Rightarrow \cosh^2 A \equiv \frac{1}{2}(\cosh 2A + 1)$

$\therefore \frac{1}{4}\int \left(\frac{1}{2}(\cosh 4u + 1) - 2\cosh 2u + 1\right)\mathrm{d}u$

$= \frac{1}{4}\int \left(\frac{1}{2}\cosh 4u - 2\cosh 2u + \frac{3}{2}\right)\mathrm{d}u$

$= \frac{1}{4}\left(\frac{1}{8}\sinh 4u - \sinh 2u + \frac{3}{2}u\right) + c$

$= \frac{1}{32}\sinh 4u - \frac{1}{4}\sinh 2u + \frac{3}{8}u + c$

$\sinh 2u = 2\sinh u \cosh u = 2\sqrt{\cosh^2 u - 1} \times \cosh u = 2\sqrt{x^2-1} \times x = 2x\sqrt{x^2-1}$

$\cosh 2u = 2\cosh^2 u - 1 = 2x^2 - 1$

$\sinh 4u = 2\sinh 2u \cosh 2u = 2 \times 2x\sqrt{x^2-1} \times (2x^2-1)$

$\therefore \int (x^2-1)^{\frac{3}{2}}\,\mathrm{d}x = \frac{1}{32} \times 4x(2x^2-1)\sqrt{x^2-1} - \frac{1}{4} \times 2x\sqrt{x^2-1} + \frac{3}{8} \times \operatorname{arcosh} x + c$

$= \frac{1}{8}x(2x^2-1)\sqrt{x^2-1} - \frac{1}{2}x\sqrt{x^2-1} + \frac{3}{8}\operatorname{arcosh} x + c$

$\therefore \frac{1}{8}x(2x^2-1)\sqrt{x^2-1} - \frac{1}{2}x\sqrt{x^2-1} + \frac{3}{8}\operatorname{arcosh} x + c$ or $\frac{1}{8}x(2x^2-1)\sqrt{x^2-1} - \frac{1}{2}x\sqrt{x^2-1} + \frac{3}{8}\ln\left(x + \sqrt{x^2-1}\right) + c$